

UCCM 1353

BASIC ALGEBRA

VECTORS

1.1 Basic Definition - Coordinate Systems

- Two dimensional plane
-any point in the plane with an ordered pair (a, b) of real numbers, a is x -coordinate, b is y - coordinate.
- Three dimensional plane
-any point in the plane with an ordered triple (a, b, c) of real numbers, a is x -coordinate, b is y - coordinate, c is z - coordinate.

1.1 Basic Definition - Coordinate Systems (Cont')

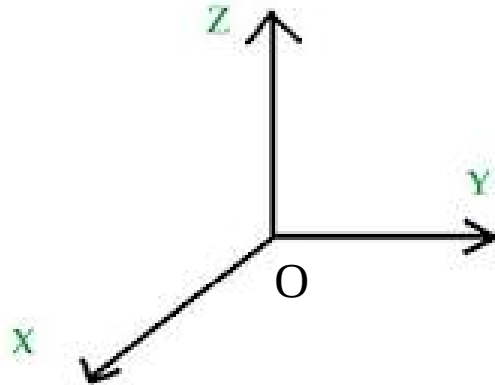
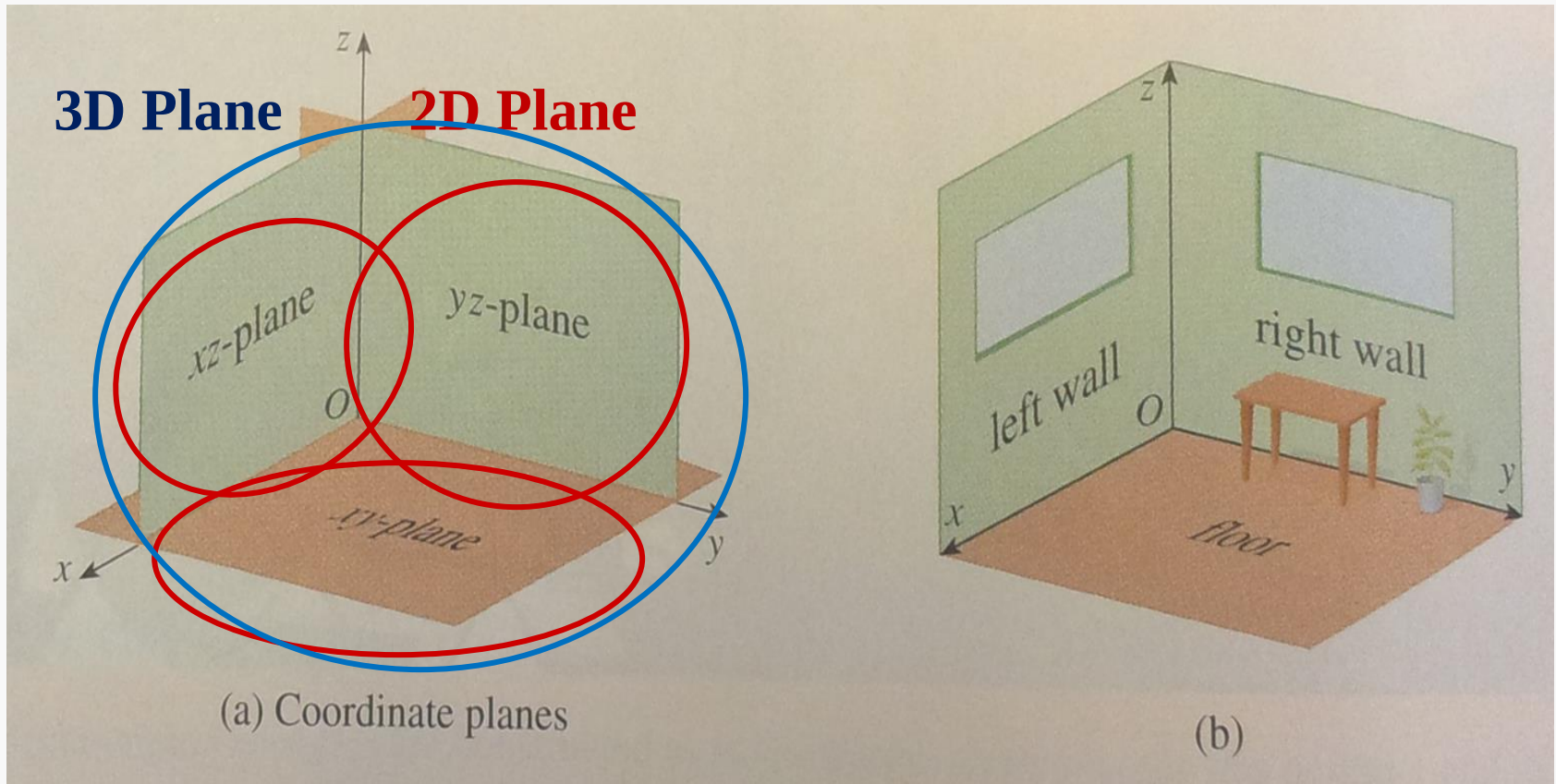


Figure 1: Coordinate axes for three dimensional plane.

- Choose a fix point O (origin) and the three directed line through O that are perpendicular to each other, called the coordinate axes.
- x and y - axes as being the horizontal and z-axis being vertical.

Coordinate Planes



Coordinate Planes

Let P be any point in the space, we represent the point P by the ordered triple (a, b, c) of real numbers and a , b and c are the coordinates of P .

- a – the x - coordinate
- b – the y - coordinate
- c – the z - coordinate
- To locate the point $P(a, b, c)$, start at the origin O and move a units along the x -axis, b units parallel to y -axis, c units parallel to the z -axis.

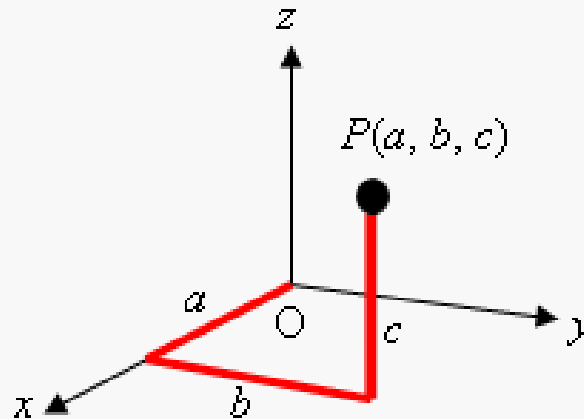


Figure : Coordinate of P .

Coordinate Planes

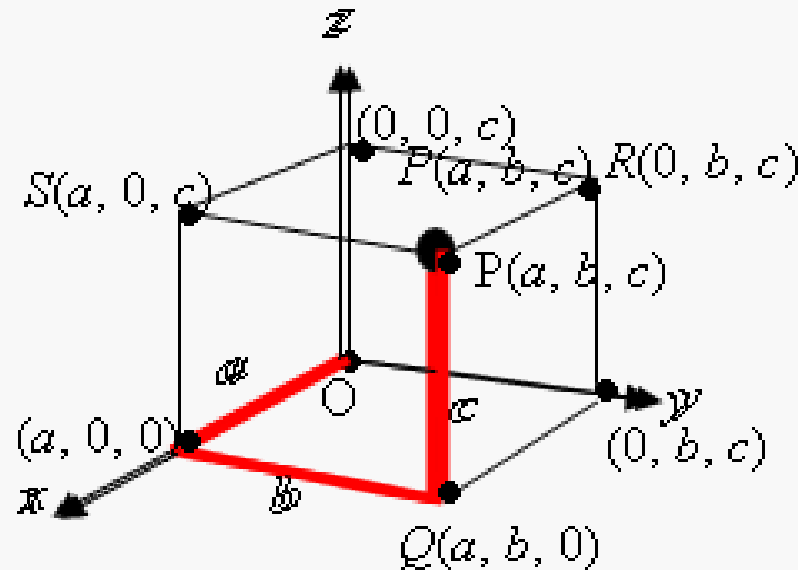


Figure: Coordinate of P determines a rectangular box.

- Drop a perpendicular from P to xy -plane, we get a point Q with coordinates $(a, b, 0)$, called the projection of P onto xy -plane.
- Similarly, $R(0, b, c)$ and $S(a, 0, c)$ are projections of P onto yz -plane and xz -plane, respectively.
- The Cartesian product $\mathcal{R} \times \mathcal{R} \times \mathcal{R} = \{(x, y, z) | x, y, z \in \mathcal{R}\}$ is the set of all ordered triples of real numbers and is denoted by $\mathcal{R}^3$.

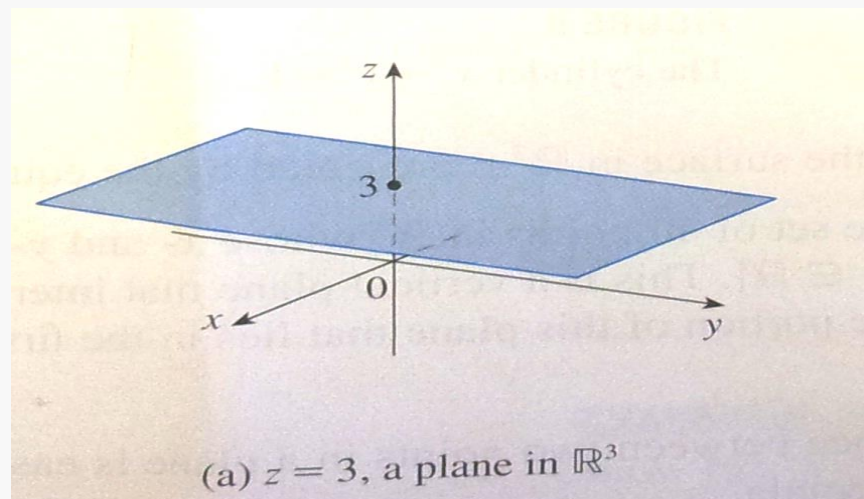
Example 1.1 (Graphing equations)

What surface in $\mathbb{R}^3$ are represented by the following equations?

(a) $z = 3$

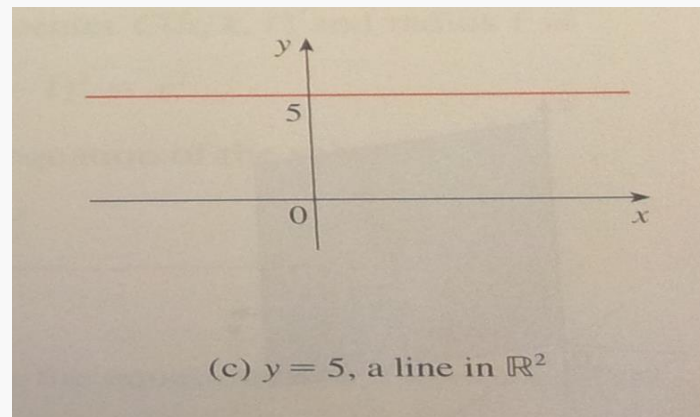
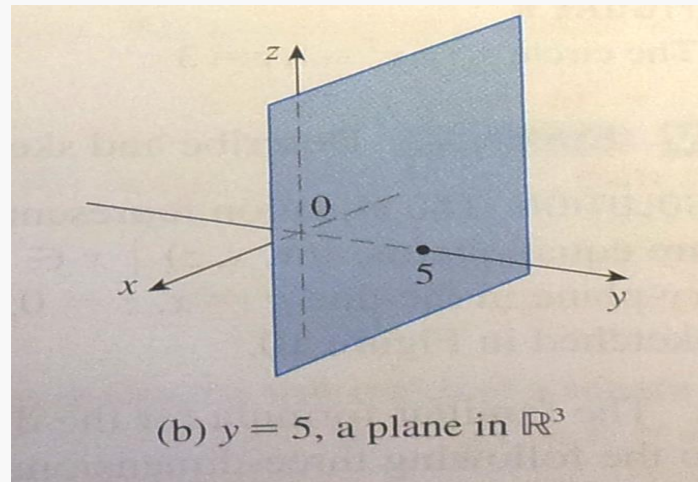
(b) $y = 5$

(a) The equation $z = 3$ represents the set which is the set of all points in whose z -coordinate is 3. This is horizontal plane that is parallel to the xy -plane and three units above it.



Example 1.1 (Graphing equations) (Cont')

(b) The equation $y = 5$ represents the set of all points in whose y -coordinate is 5. This is vertical plane that is parallel to the xz -plane and three units above it.



Length/Distance Formula in Three Dimensions

►► The distance $|P_1P_2|$ between the points $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ is

$$|P_1P_2| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

►► The distance from the point $P(2, -1, 7)$ to the point $Q(1, -3, 5)$ is

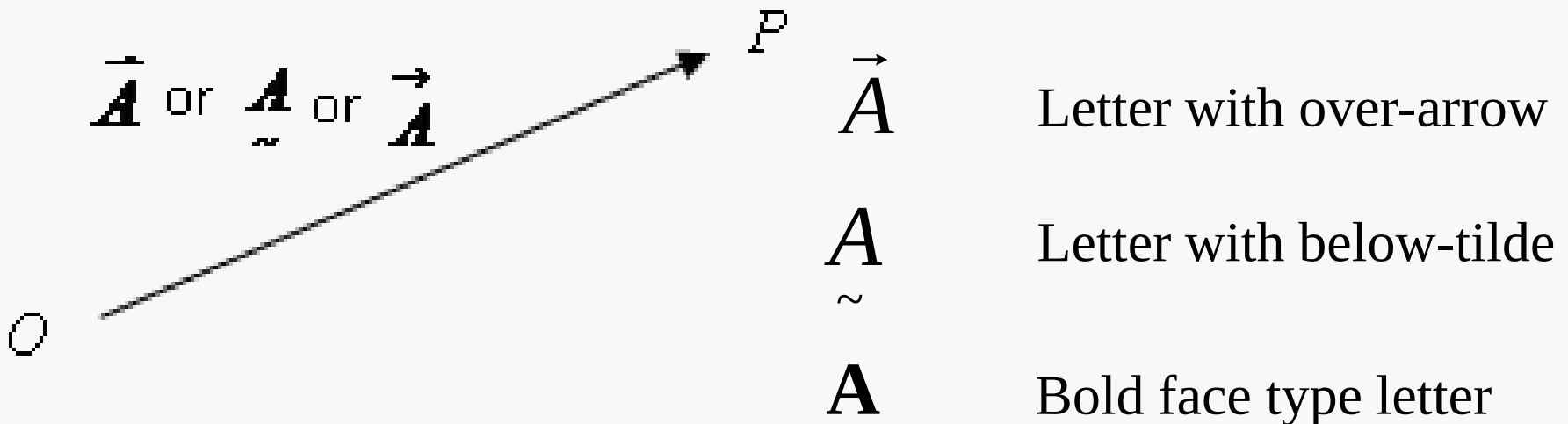
$$\begin{aligned} |\overrightarrow{PQ}| &= \sqrt{(1-2)^2 + (-3-(-1))^2 + (5-7)^2} \\ &= \sqrt{9} = 3 \end{aligned}$$

Vector

- ♣ Physical quantities (measurable) can be categorized into vector and scalar.
- ♣ A **scalar** is a quantity having magnitude but no direction, e.g. mass, length, time, temperature, and real number.
- ♣ A **vector** is a quantity having both magnitude and direction, e.g. displacement, velocity, force, and acceleration.

Vector (Cont')

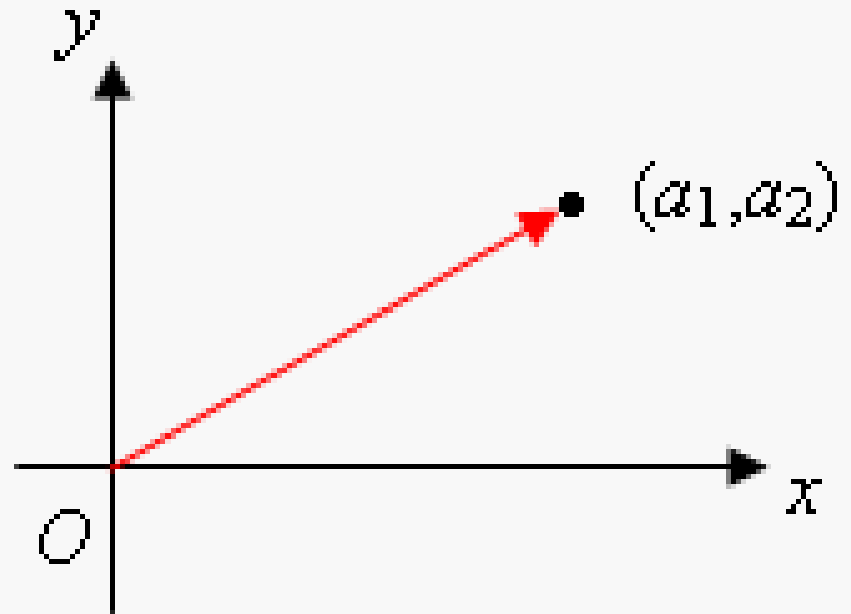
- ♣ represented by arrow OP . which defines the direction while the magnitude of the vector is indicated by the length of the arrow. The tail end O is called origin or initial point of the vector; and the head P is called the terminal point.



2D Vector

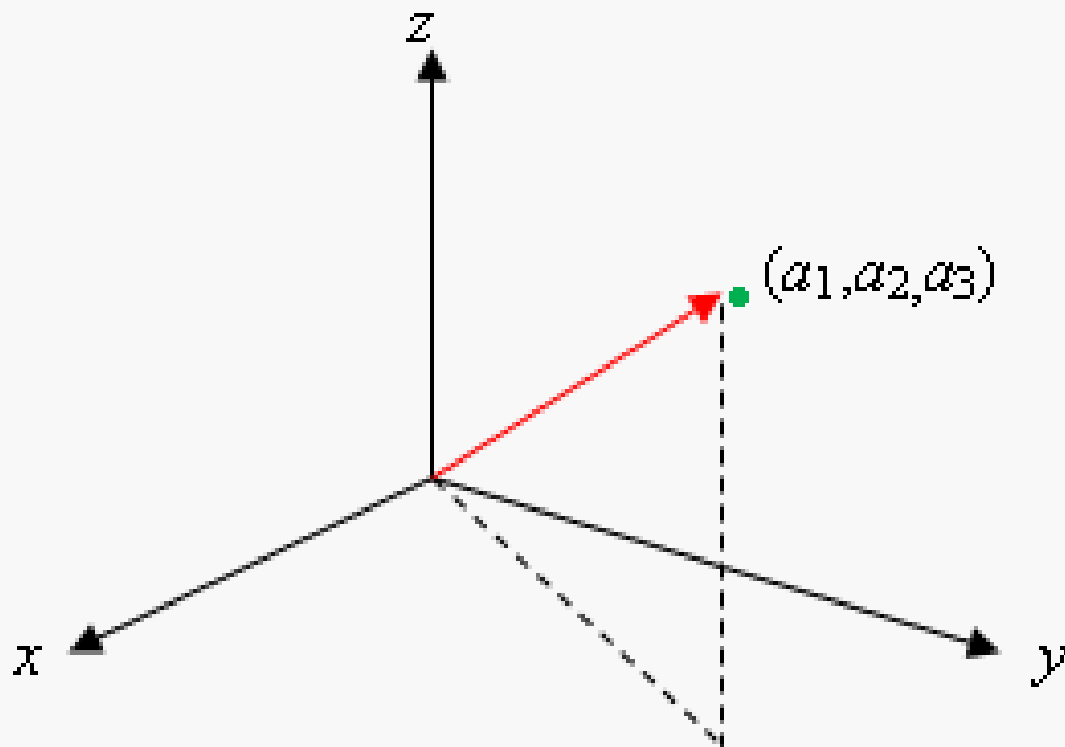
- Involve only two direction coordinate planes:

xy -, xz - or yz -plane



3D Vector

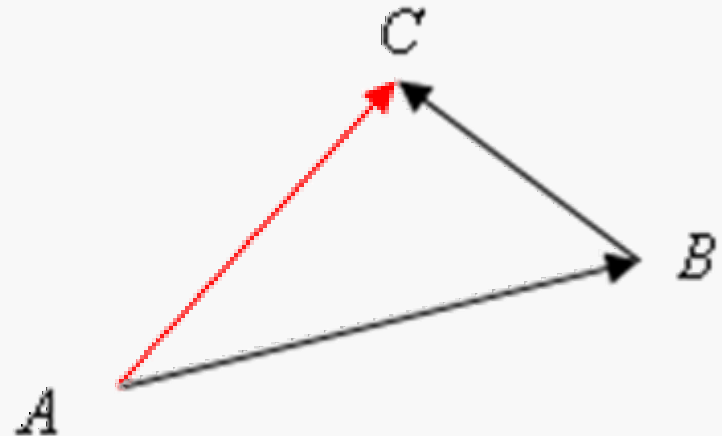
⊖ Involve all three direction coordinate
planes:
xyz-plane



Combination of Vectors

- ▶ Suppose a particle moves from A to B:
so its displacement vector is $\overrightarrow{AB}$.
- ▶ Then the particle changes direction and moves from B to C:
displacement vector $\overrightarrow{BC}$.
- ▶ The combined effect of these displacements:
the particle has moved from A to C.
- ▶ The resulting displacement vector $\overrightarrow{AC}$ is called the sum of $\overrightarrow{AB}$ and $\overrightarrow{BC}$ can be written as

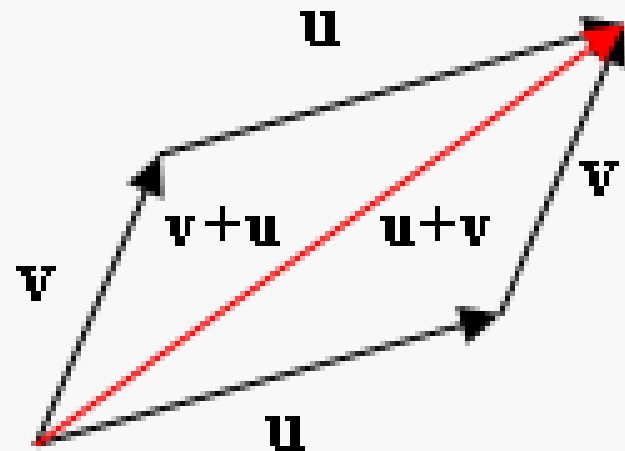
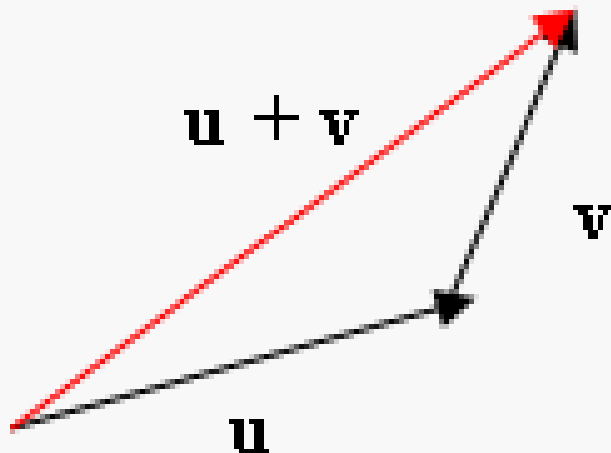
$$\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC}$$



Combination of Vectors (Cont')

Vector **Addition**:

If $\mathbf{u}$ and $\mathbf{v}$ are vectors positioned so the initial point of $\mathbf{v}$ is at the terminal point of $\mathbf{u}$, then the **sum** ($\mathbf{u} + \mathbf{v}$) is the vector from the initial point of $\mathbf{u}$ to the terminal point of $\mathbf{v}$.

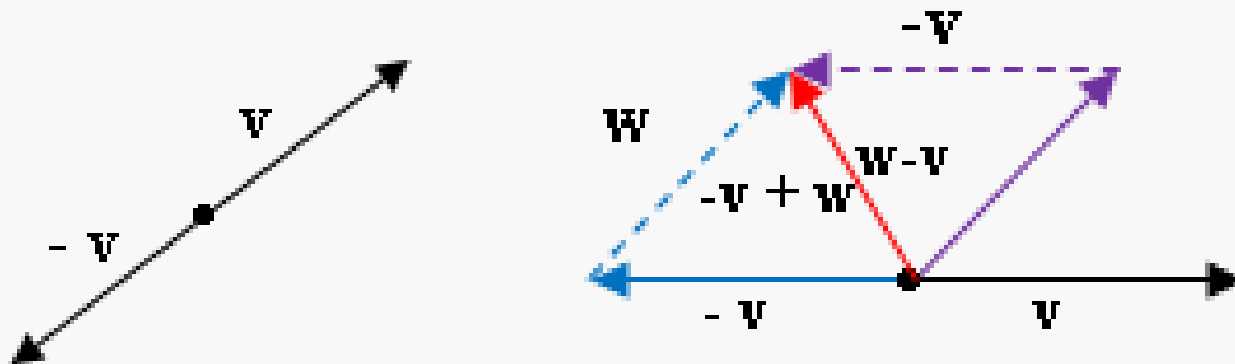


Combination of Vectors (Cont')

Vector **Subtraction**:

The **negative** of a vector $\mathbf{v}$, denoted by $-\mathbf{v}$, is the vector that has the same length as $\mathbf{v}$ but is oppositely directed, and the **difference** of $\mathbf{v}$ from $\mathbf{w}$, denoted by $\mathbf{w} - \mathbf{v}$, is taken to be the sum

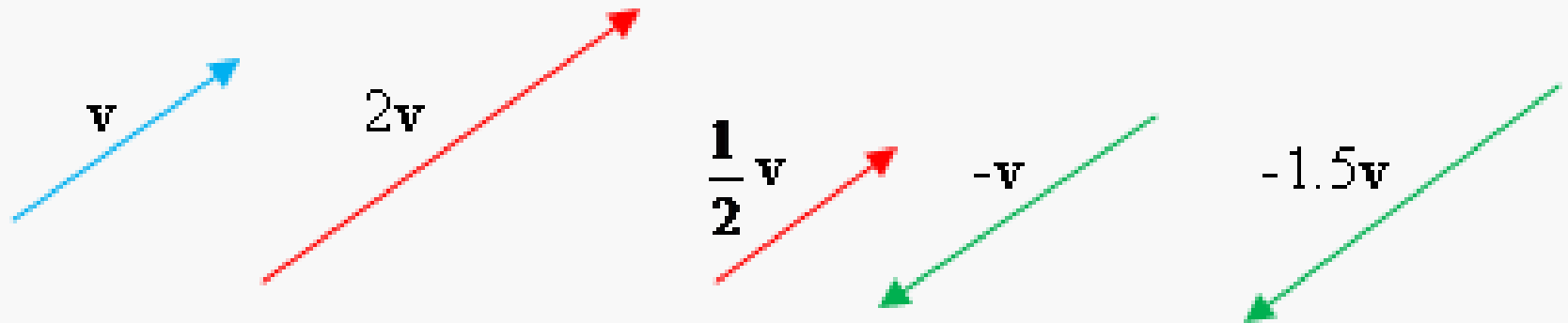
$$\mathbf{w} - \mathbf{v} = \mathbf{w} + (-\mathbf{v})$$



Combination of Vectors (Cont')

Scalar **Multiplication**:

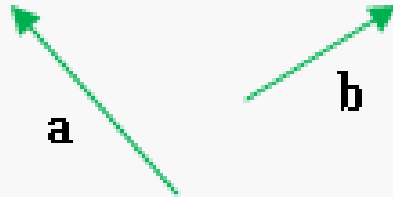
If c is scalar and $\mathbf{v}$ is a vector, then the **scalar multiple** $c\mathbf{v}$ is the vector whose length is times the length of $\mathbf{v}$ and whose direction is the same as $\mathbf{v}$ if $c > 0$ and is opposite to $\mathbf{v}$ if $c < 0$. If $c = 0$ or $\mathbf{v} = 0$, then $c\mathbf{v} = 0$.



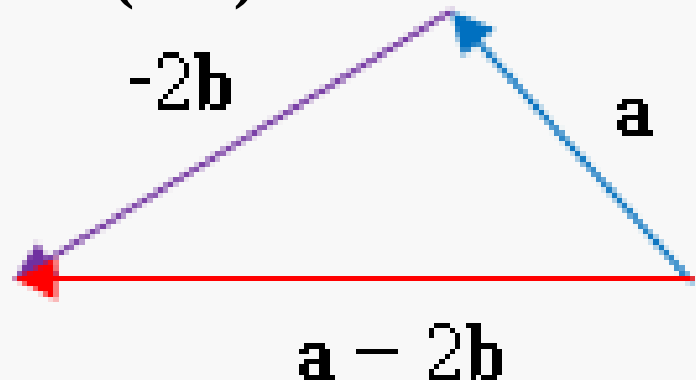
Combination of Vectors (Cont')

Example:

If **a** and **b** are the vectors shown as below, draw $\mathbf{a} - 2\mathbf{b}$



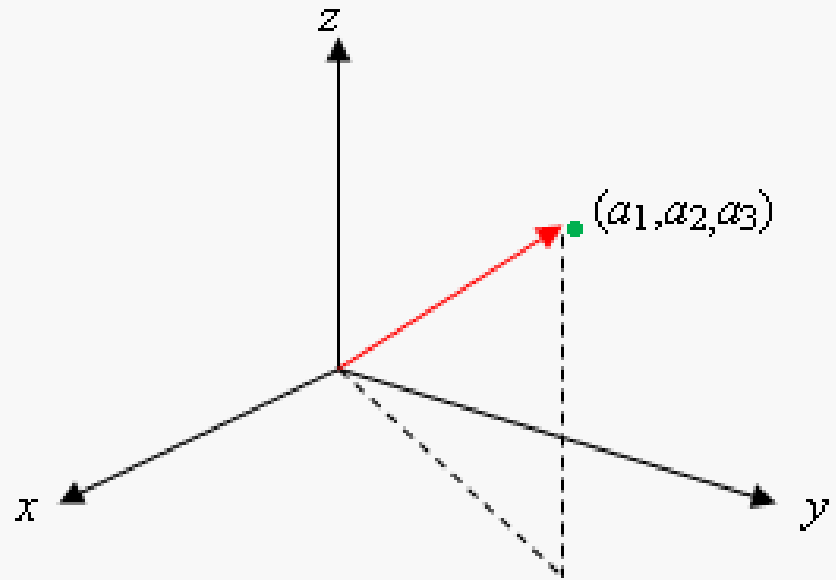
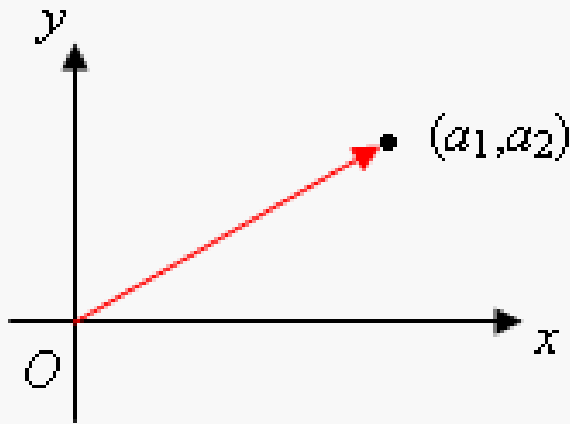
First draw the vector $-2\mathbf{b}$ pointing in the direction opposite to **b** and twice as long. We place it with its tail at the tip of **a** and then use the Triangle Law to draw $\mathbf{a} + (-2\mathbf{b})$



Components

Components of vector **a**: $\langle a_1, a_2 \rangle$ or $\langle a_1, a_2, a_3 \rangle$

The coordinates of a vector from the origin **O** of the coordinate system where terminal point of **a** has coordinates of the form (a_1, a_2) or (a_1, a_2, a_3) .



Components (Cont')

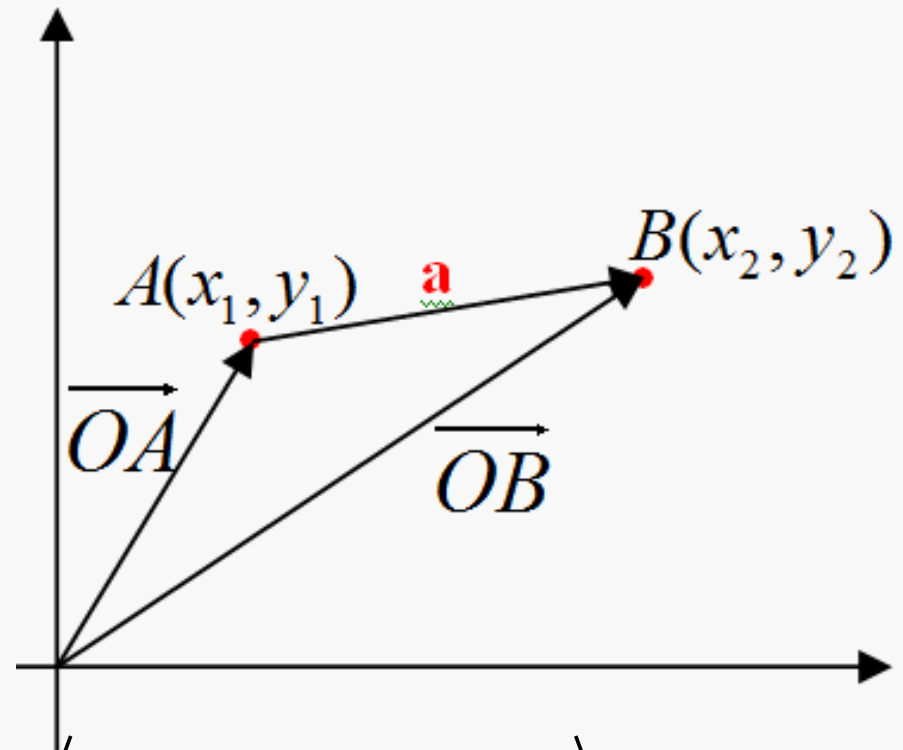
$$\mathbf{a} = \langle x_2 - x_1, y_2 - y_1 \rangle$$

$$\overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB}$$

$$= -\overrightarrow{OA} + \overrightarrow{OB}$$

$$= \overrightarrow{OB} - \overrightarrow{OA}$$

$$= \langle x_2, y_2 \rangle - \langle x_1, y_1 \rangle = \langle x_2 - x_1, y_2 - y_1 \rangle$$



Components (Cont')

Example:

Find the vector represented by the directed line segment with initial point $A(2, -3, 4)$ and terminal point $B(-2, 1, 1)$.

$$\overrightarrow{\mathbf{AB}} = \langle -2 - 2, 1 - (-3), 1 - 4 \rangle = \langle -4, 4, -3 \rangle$$

or

$$\overrightarrow{\mathbf{AB}} = \begin{pmatrix} (-2) - 2 \\ 1 - (-3) \\ 1 - 4 \end{pmatrix} = \begin{pmatrix} -4 \\ 4 \\ -3 \end{pmatrix}$$

Norm

♣ The **magnitude** or **length** of a vector **v**

♣ denoted by the symbol $|\mathbf{v}|$ or $\|\mathbf{v}\|$

♣ $|\mathbf{a}| = \sqrt{a_1^2 + a_2^2}$ or $\sqrt{a_1^2 + a_2^2 + a_3^2}$

Operation Vectors

3

If $\mathbf{a} = \langle a_1, a_2 \rangle$ and $\mathbf{b} = \langle b_1, b_2 \rangle$, then

$$\mathbf{a} + \mathbf{b} = \langle a_1 + b_1, a_2 + b_2 \rangle$$

$$\mathbf{a} - \mathbf{b} = \langle a_1 - b_1, a_2 - b_2 \rangle$$

$$c\mathbf{a} = \langle ca_1, ca_2 \rangle$$

Similarly, for three-dimensional vectors,

$$\langle a_1, a_2, a_3 \rangle + \langle b_1, b_2, b_3 \rangle = \langle a_1 + b_1, a_2 + b_2, a_3 + b_3 \rangle$$

$$\langle a_1, a_2, a_3 \rangle - \langle b_1, b_2, b_3 \rangle = \langle a_1 - b_1, a_2 - b_2, a_3 - b_3 \rangle$$

$$c\langle a_1, a_2, a_3 \rangle = \langle ca_1, ca_2, ca_3 \rangle$$

Example 1.6(Operation Vectors)

If $\mathbf{a} = \langle 4, 0, 3 \rangle$ and $\mathbf{b} = \langle -2, 1, 5 \rangle$ find $|\mathbf{a}|$ and the vectors $\mathbf{a} + \mathbf{b}$, $\mathbf{a} - \mathbf{b}$, $3\mathbf{b}$, and $2\mathbf{a} + 5\mathbf{b}$

1. $|\mathbf{a}| = \sqrt{4^2 + 0^2 + 3^2} = \sqrt{25} = 5$
2. $\mathbf{a} + \mathbf{b} = \langle 4, 0, 3 \rangle + \langle -2, 1, 5 \rangle = \langle 2, 1, 8 \rangle$
3. $\mathbf{a} - \mathbf{b} = \langle 4, 0, 3 \rangle - \langle -2, 1, 5 \rangle = \langle 6, -1, -2 \rangle$
4. $3\mathbf{b} = 3\langle -2, 1, 5 \rangle = \langle -6, 3, 15 \rangle$
5. $2\mathbf{a} + 5\mathbf{b} = 2\langle 4, 0, 3 \rangle + 5\langle -2, 1, 5 \rangle = \langle -2, 5, 31 \rangle$

Properties of Vectors

- Commutative Law $\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}$
- Associative Law $\mathbf{a} + (\mathbf{b} + \mathbf{c}) = (\mathbf{a} + \mathbf{b}) + \mathbf{c}$
 $(cd) \mathbf{a} = c (d \mathbf{a})$
- Distributive Law $c (\mathbf{a} + \mathbf{b}) = c \mathbf{a} + c \mathbf{b}$
 $(c + d) \mathbf{a} = c \mathbf{a} + d \mathbf{a}$
- $\mathbf{a} + \mathbf{0} = \mathbf{a}$
- $\mathbf{a} + (-\mathbf{a}) = \mathbf{0}$
- $1 \mathbf{a} = \mathbf{a}$

The Standard Unit Vectors

- When a rectangular coordinate system is introduced in R^2 and R^3 , the unit vectors in the positive directions of coordinate axes are called the standard unit vectors.

- In R^2 these vectors are denoted by

$$i = \langle 1, 0 \rangle \quad j = \langle 0, 1 \rangle$$

- In R^3 :

$$\mathbf{i} = \langle 1, 0, 0 \rangle \quad \mathbf{j} = \langle 0, 1, 0 \rangle \quad \mathbf{k} = \langle 0, 0, 1 \rangle$$

The Standard Unit Vectors (Con't)

- Then **i**, **j**, **k** are vectors that have length 1 and point in the directions of the positive x -, y -, and z -axes and similarly in two dimensions.

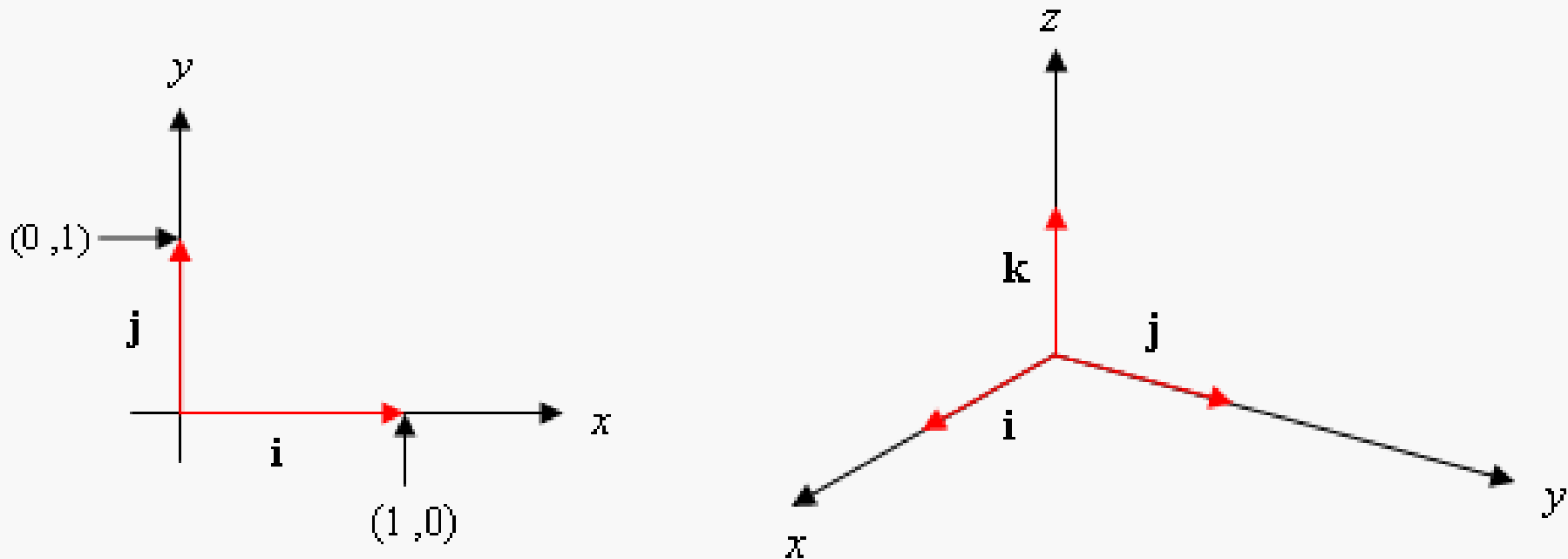


Figure : Standard basis vectors in R^2 and R^3 .

The Standard Unit Vectors (Con't)

If $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$, then we can write

$$\begin{aligned}\mathbf{a} &= \langle a_1, a_2, a_3 \rangle = \langle a_1, 0, 0 \rangle + \langle 0, a_2, 0 \rangle + \langle 0, 0, a_3 \rangle \\ &= a_1 \langle 1, 0, 0 \rangle + a_2 \langle 0, 1, 0 \rangle + a_3 \langle 0, 0, 1 \rangle\end{aligned}$$

$$\mathbf{a} = a_1 \mathbf{i} + a_2 \mathbf{j} + a_3 \mathbf{k}$$

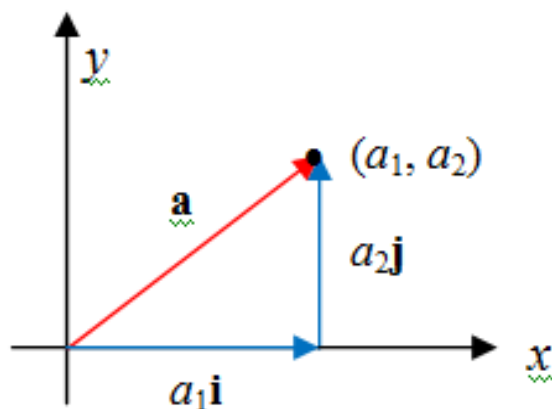


Figure 17(a): $\mathbf{a} = a_1 \mathbf{i} + a_2 \mathbf{j}$

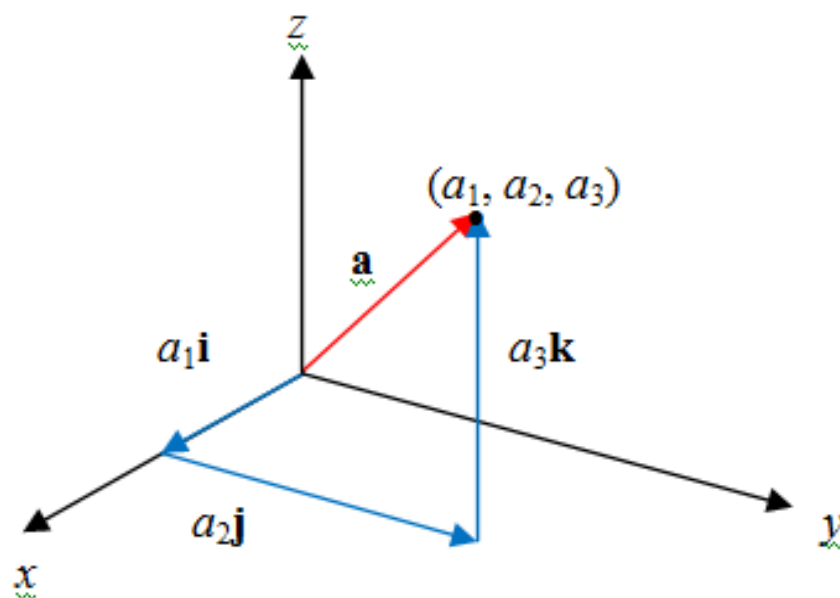


Figure 17 (b): $\mathbf{a} = a_1 \mathbf{i} + a_2 \mathbf{j} + a_3 \mathbf{k}$

Thus any vector in can be expressed in term of the standard basis vectors $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$.

The Standard Unit Vectors (Con't)

For instance,

$$\langle 1, -2, 6 \rangle = \mathbf{i} - 2\mathbf{j} + 6\mathbf{k}$$

Similarly, in two dimensions, we can write

$$\mathbf{a} = \langle a_1, a_2 \rangle = a_1\mathbf{i} + a_2\mathbf{j}$$

Example 1.7:

If $\mathbf{a} = \mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$ and $\mathbf{b} = 4\mathbf{i} + 7\mathbf{k}$, express the vector $2\mathbf{a} + 3\mathbf{b}$ in terms of $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$.

Solution:

$$2\mathbf{a} + 3\mathbf{b} = 2(\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}) + 3(4\mathbf{i} + 7\mathbf{k}) = 2\mathbf{i} + 4\mathbf{j} - 6\mathbf{k} + 12\mathbf{i} + 21\mathbf{k} = 14\mathbf{i} + 4\mathbf{j} + 15\mathbf{k}$$

Unit Vector

- A vector whose length = 1. For instance, **i**, **j**, **k** all are unit vectors. In general, if **a** $\neq$ 0, then the unit vector that has the same direction as **a** is

$$\hat{\mathbf{u}} = \frac{\mathbf{a}}{\|\mathbf{a}\|}$$

Unit Vector (Cont')

Example 1.8

Find the unit vector in the direction of the vector $2\mathbf{i} - \mathbf{j} - 2\mathbf{k}$.

$$\|\mathbf{a}\| = \sqrt{2^2 + (-1)^2 + (-2)^2} = \sqrt{9} = 3$$

$$\hat{\mathbf{u}} = \frac{\mathbf{a}}{\|\mathbf{a}\|} = \frac{2\mathbf{i} - \mathbf{j} - 2\mathbf{k}}{3} = \frac{2}{3}\mathbf{i} - \frac{1}{3}\mathbf{j} - \frac{2}{3}\mathbf{k}$$

Dot Product

•If $\mathbf{a} = \langle a_1, a_2, \dots, a_n \rangle$ and $\mathbf{b} = \langle b_1, b_2, \dots, b_n \rangle$ are vectors in R^n , then the dot product of $\mathbf{a}$ and $\mathbf{b}$ is denoted by $\mathbf{a} \cdot \mathbf{b}$ and is defined by

$$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + \dots + a_nb_n$$

Dot Product (Cont')

Example 1.9

1. $\langle 2, 4 \rangle \cdot \langle 3, -1 \rangle$

$$= 2(3) + 4(-1) = 2$$

2. $\langle -1, 7, 4 \rangle \cdot \langle 6, 2, -0.5 \rangle$

$$= (-1)(6) + 7(2) + 4(-0.5) = 6$$

3. $(\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}) \cdot (2\mathbf{j} - \mathbf{k})$

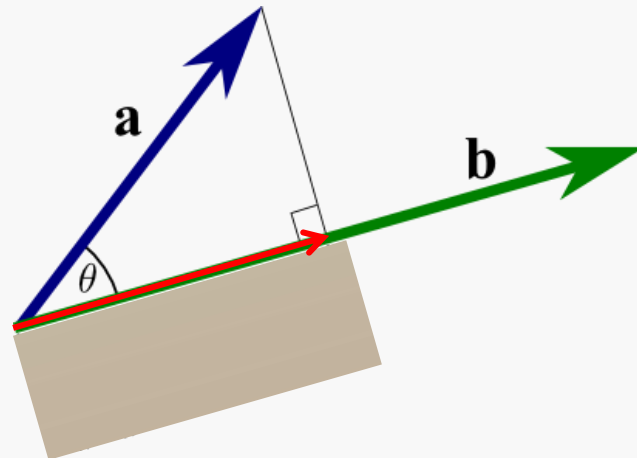
$$= 1(0) + 2(2) + (-3)(-1) = 7$$

Dot Product (Cont')

If **a** and **b** are nonzero vectors, and θ is the angle between **a** and **b**, then the dot product of **a** and **b** is defined as

$$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta$$

θ is the smaller angle between the vectors when they are drawn with the same initial point. If either **a** or **b** is 0, then $\mathbf{a} \cdot \mathbf{b} = 0$.



Dot Product (Cont')

- Two nonzero vectors **a** and **b** are called **perpendicular** or **orthogonal** if the angle between them is 90° or $\pi/2$. Then

$$\mathbf{a} \bullet \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos(\pi/2) = 0$$

Conversely if $\mathbf{a} \bullet \mathbf{b} = 0$, then $\theta = \pi/2$.

Example 1.10

Show that $2\mathbf{i} + 2\mathbf{j} - \mathbf{k}$ is $\perp$ to $5\mathbf{i} - 4\mathbf{j} + 2\mathbf{k}$

$$(2\mathbf{i} + 2\mathbf{j} - \mathbf{k}) \bullet (5\mathbf{i} - 4\mathbf{j} + 2\mathbf{k})$$

$$= 2(5) + 2(-4) + (-1)(2)$$

$$= 0$$

Dot Product (Cont')

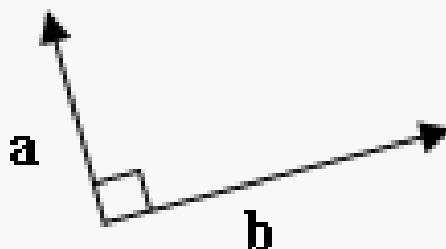
$$\Rightarrow \theta < \pi/2, \mathbf{a} \cdot \mathbf{b} > 0$$

$$\Rightarrow \theta > \pi/2, \mathbf{a} \cdot \mathbf{b} < 0$$

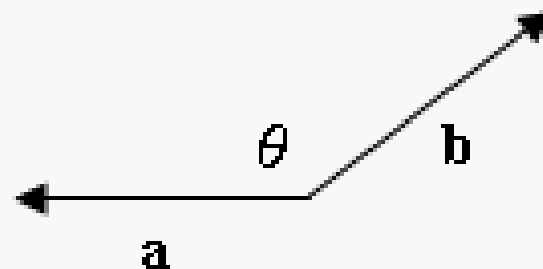
θ acute
 $0 < \theta < 90^\circ$



$\theta = \pi/2$



θ obtuse
 $90^\circ < \theta < 180^\circ$



Dot Product (Cont')

Example 1.11:

Find the angle between $\mathbf{a} = \langle 2, 2, -1 \rangle$ and $\mathbf{b} = \langle 5, -3, 2 \rangle$.

$$\mathbf{a} \bullet \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta = 0$$

$$2(5) + 2(-3) + (-1)(2) = \sqrt{2^2 + 2^2 + (-1)^2} \sqrt{5^2 + (-3)^2 + 2^2} \cos \theta$$

$$2 = 3\sqrt{38} \cos \theta$$

$$\theta = \cos^{-1} \left(\frac{2}{3\sqrt{38}} \right) \approx 1.46 \text{ rad (or } 84^\circ)$$

Dot Product (Cont')

Properties:

a, **b** and **c** are vectors in V_3 and k is a scalar, then

i. $\mathbf{a} \bullet \mathbf{a} = \|\mathbf{a}\|^2$

ii. $\mathbf{a} \bullet \mathbf{b} = \mathbf{b} \bullet \mathbf{a}$

iii. $\mathbf{a} \bullet (\mathbf{b} + \mathbf{c}) = \mathbf{a} \bullet \mathbf{b} + \mathbf{a} \bullet \mathbf{c}$

iv. $(k\mathbf{a}) \bullet \mathbf{b} = k(\mathbf{a} \bullet \mathbf{b}) = \mathbf{a} \bullet (k\mathbf{b})$

v. $\mathbf{0} \bullet \mathbf{a} = 0$

Cross Product

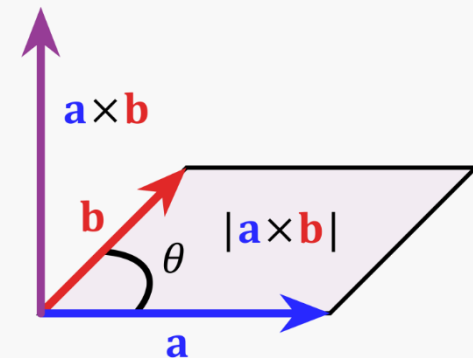
- The cross product $\mathbf{a} \times \mathbf{b}$ of two vectors $\mathbf{a}$ and $\mathbf{b}$, unlike the dot product, is a vector.
- Note that $\mathbf{a} \times \mathbf{b}$ is defined only when $\mathbf{a}$ and $\mathbf{b}$ are three-dimensional (3-D) vectors.

Definition If $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$ are vectors in 3-space, then the cross product is the vector defined by

$$\mathbf{a} \times \mathbf{b} = \langle a_2 b_3 - a_3 b_2, \quad a_3 b_1 - a_1 b_3, \quad a_1 b_2 - a_2 b_1 \rangle$$

or in determinant notation:
$$= \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \mathbf{k}$$

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \mathbf{k}$$



Cross Product (Cont')

Example 1.12:

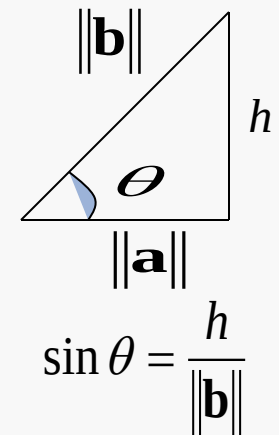
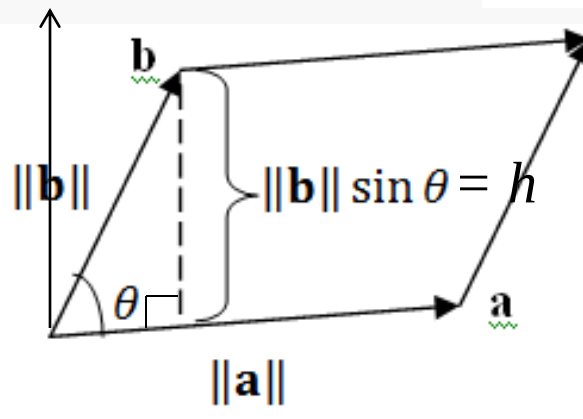
Find $\mathbf{u} \times \mathbf{v}$ where $\mathbf{u} = \langle 1, 2, -2 \rangle$ and $\mathbf{v} = \langle 3, 0, 1 \rangle$

$$\begin{aligned}\mathbf{u} \times \mathbf{v} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & -2 \\ 3 & 0 & 1 \end{vmatrix} = \begin{vmatrix} 2 & -2 \\ 0 & 1 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 1 & -2 \\ 3 & 1 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 1 & 2 \\ 3 & 0 \end{vmatrix} \mathbf{k} \\ &= \langle 2 - 0, -[1 - (-6)], 0 - 6 \rangle \\ &= \langle 2, -7, -6 \rangle\end{aligned}$$

Geometric Interpretation of Cross Product

- If $\mathbf{u}$ and $\mathbf{v}$ are vectors in 3-space, then the norm of $\mathbf{u} \times \mathbf{v}$ has a useful geometric interpretation. Hence we have

$$\|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\| \|\mathbf{b}\| \sin \theta, \text{ for } 0 \leq \theta \leq \pi$$



$$h = \|\mathbf{b}\| \sin \theta$$

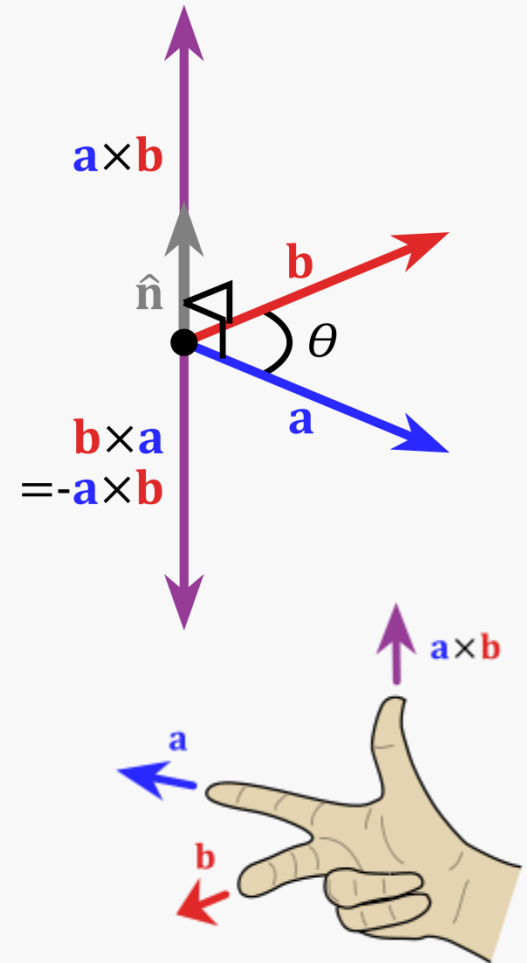
- But $\|\mathbf{b}\| \sin \theta$ is the altitude (height) of the parallelogram determined by $\mathbf{a}$ and $\mathbf{b}$ (figure above). Thus the area A of this parallelogram is given by

$$A = (\text{base})(\text{altitude}) = \|\mathbf{a}\| \|\mathbf{b}\| \sin \theta = \|\mathbf{a} \times \mathbf{b}\|$$

Cross Product (Properties)

⚡ Properties

- i. $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a} = -(-\mathbf{a} \times \mathbf{b})$
- ii. $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) + (\mathbf{a} \times \mathbf{c})$
- iii. $(\mathbf{a} + \mathbf{b}) \times \mathbf{c} = (\mathbf{a} \times \mathbf{c}) + (\mathbf{b} \times \mathbf{c})$
- iv. $k(\mathbf{a} \times \mathbf{b}) = k\mathbf{a} \times \mathbf{b} = \mathbf{a} \times k\mathbf{b}$
- v. $\mathbf{a} \times \mathbf{0} = \mathbf{0} \times \mathbf{a} = \mathbf{0}$
- vi. $\mathbf{a} \times \mathbf{a} = \mathbf{0}$



Cross Product (Cont')

⚡ Properties

$$\mathbf{i} \times \mathbf{i} = 0$$

$$\mathbf{j} \times \mathbf{j} = 0$$

$$\mathbf{k} \times \mathbf{k} = 0$$

$$\mathbf{i} \times \mathbf{j} = \mathbf{k}$$

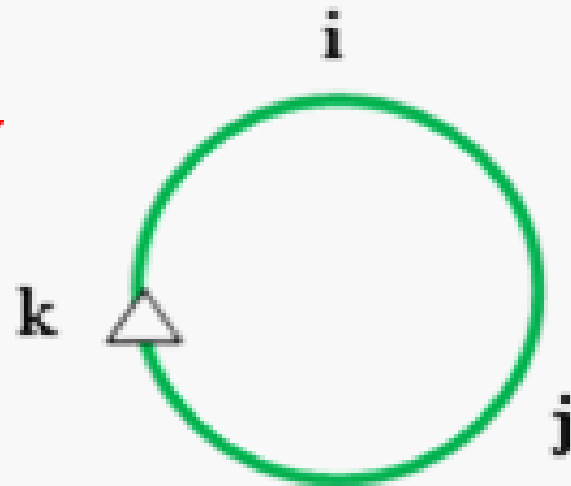
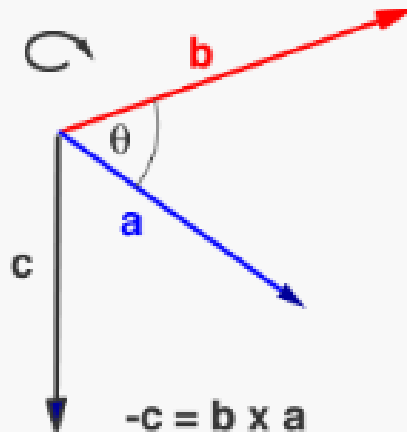
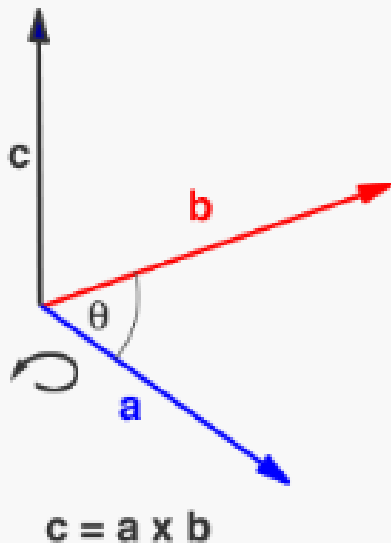
$$\mathbf{j} \times \mathbf{k} = \mathbf{i}$$

$$\mathbf{k} \times \mathbf{i} = \mathbf{j}$$

$$\mathbf{j} \times \mathbf{i} = -\mathbf{k}$$

$$\mathbf{k} \times \mathbf{j} = -\mathbf{i}$$

$$\mathbf{i} \times \mathbf{k} = -\mathbf{j}$$



Area of Parallelogram

✦ If $\mathbf{a}$ and $\mathbf{b}$ are vectors in 3D, then $\|\mathbf{a} \times \mathbf{b}\|$ is equal to the area of the parallelogram determined by $\mathbf{a}$ and $\mathbf{b}$.

Example 1.13:

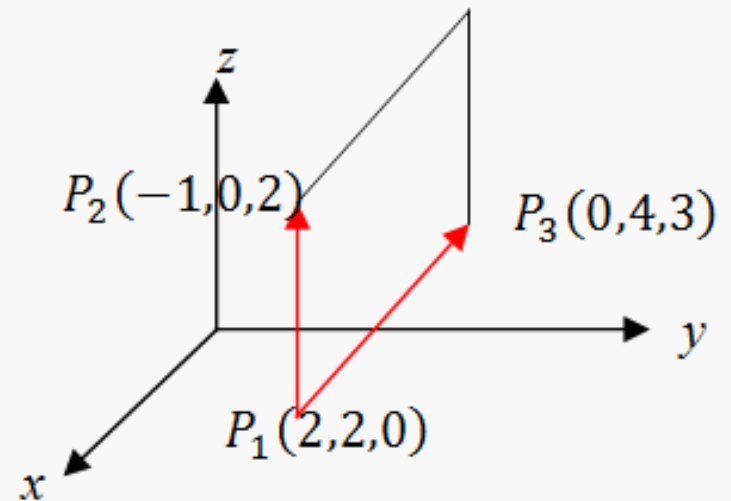
Find the area of the triangle determined by the points $P_1(2, 2, 0)$, $P_2(-1, 0, 2)$ and $P_3(0, 4, 3)$.

$$\overrightarrow{P_1P_2} = P_1O + OP_2 = \langle -3, -2, 2 \rangle$$

$$\overrightarrow{P_1P_3} = P_1O + OP_3 = \langle -2, 2, 3 \rangle$$

$$\overrightarrow{P_1P_2} \times \overrightarrow{P_1P_3} = \langle -10, 5, -10 \rangle$$

$$\text{Area, } A = \frac{1}{2} \left\| \overrightarrow{P_1P_2} \times \overrightarrow{P_1P_3} \right\| = \frac{1}{2} (15) = \frac{15}{2}$$



Relationships Involving Cross Product and Dot Product

If $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$, $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$ and $\mathbf{c} = \langle c_1, c_2, c_3 \rangle$ are vectors in 3-space, then

- i. $\mathbf{a} \cdot (\mathbf{a} \times \mathbf{b}) = 0$
 - ii. $\mathbf{b} \cdot (\mathbf{a} \times \mathbf{b}) = 0$
- } $\mathbf{a} \times \mathbf{b}$ is perpendicular to $\mathbf{a}$ and $\mathbf{b}$
- iii. $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}$
 - iv. $(\mathbf{a} \times \mathbf{b}) \times \mathbf{c} = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{b} \cdot \mathbf{a})\mathbf{c}$

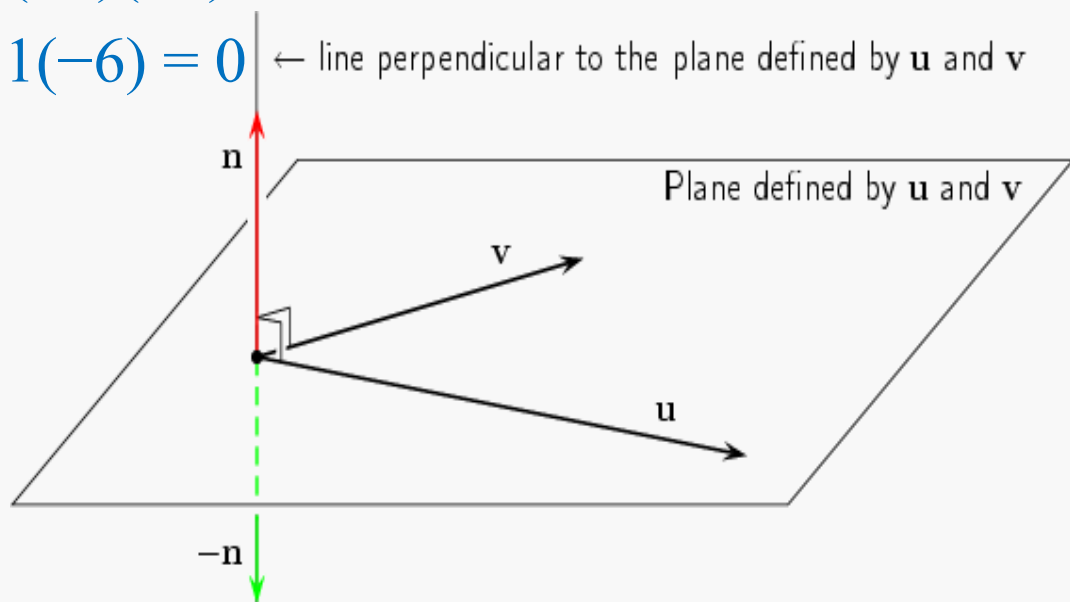
Relationships Involving Cross Product and Dot Product (Cont')

Example 1.14: Given $\mathbf{u} = \langle 1, 2, -2 \rangle$ and $\mathbf{v} = \langle 3, 0, 1 \rangle$. Show that $\mathbf{u} \times \mathbf{v}$ is perpendicular to $\mathbf{u}$ and $\mathbf{v}$.

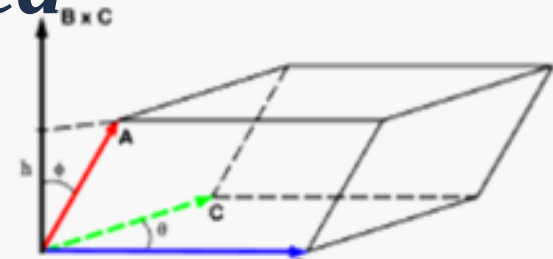
$$\mathbf{u} \times \mathbf{v} = \langle 2, -7, -6 \rangle$$

$$\mathbf{u} \cdot (\mathbf{u} \times \mathbf{v}) = 1(2) + 2(-7) + (-2)(-6) = 0$$

$$\mathbf{v} \cdot (\mathbf{u} \times \mathbf{v}) = 3(2) + 0(-7) + 1(-6) = 0$$



Volume of a Parallelepiped



Volume of the parallelepiped formed by the vectors **a**, **b** and **c** is given by the **absolute value** of their scalar triple product:

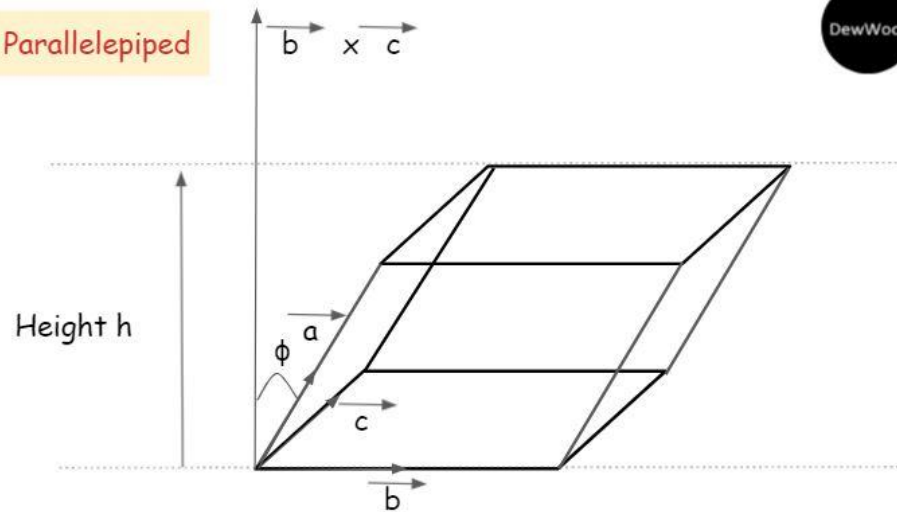
$$V = |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$$

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \text{ Determinant}$$

volume = 0, if **a, **b** and **c** all lie on the same plane (Coplanar)

Parallelepiped

DewWool



$$\begin{aligned}\text{Volume of a parallelepiped} &= \text{base area} \times \text{height} \\ &= |\vec{b} \times \vec{c}| |\vec{a}| |\cos \phi| \\ &= |\vec{b} \times \vec{c}| \cdot |\vec{a}|\end{aligned}$$

Volume of a Parallelepiped (Cont')

Example 1.17:

Find the volume of parallelepiped determined by the vectors $\mathbf{a} = \langle 1, 0, 6 \rangle$, $\mathbf{b} = \langle 2, 3, -8 \rangle$ and $\mathbf{c} = \langle 8, -5, 6 \rangle$.

$$\mathbf{b} \times \mathbf{c} = \begin{bmatrix} 2 \\ 3 \\ -8 \end{bmatrix} \times \begin{bmatrix} 8 \\ -5 \\ 6 \end{bmatrix} = \begin{bmatrix} 18 - 40 \\ -64 - 12 \\ -10 - 24 \end{bmatrix} = \begin{bmatrix} -22 \\ -76 \\ -34 \end{bmatrix}$$

$$\begin{aligned} V &= | \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) | = | \langle 1, 0, 6 \rangle \cdot \langle -22, -76, -34 \rangle | \\ &= | 1(-22) + 0(-76) + 6(-34) | \\ &= | -226 | = 226 \text{ unit}^3 \end{aligned}$$

Volume of a Parallelepiped (Cont')

Example 1.17:

Find the volume of parallelepiped determined by the vectors $\mathbf{a} = \langle 1, 0, 6 \rangle$, $\mathbf{b} = \langle 2, 3, -8 \rangle$ and $\mathbf{c} = \langle 8, -5, 6 \rangle$.

$$V = | \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) |$$

$$= \begin{vmatrix} 1 & 0 & 6 \\ 2 & 3 & -8 \\ 8 & -5 & 6 \end{vmatrix} = 1 \begin{vmatrix} 3 & -8 \\ -5 & 6 \end{vmatrix} - 0 \begin{vmatrix} 2 & -8 \\ 8 & 6 \end{vmatrix} + 6 \begin{vmatrix} 2 & 3 \\ 8 & -5 \end{vmatrix}$$

determinant

$$= | 1(18 - 40) - 0 + 6(-10 - 24) |$$

absolute

$$= | -22 + 6(-34) |$$

$$= 226 \text{ unit}^3$$

Coplanar

If the vectors $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$, $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$ and $\mathbf{c} = \langle c_1, c_2, c_3 \rangle$ have the same initial point, then they lie in the same plane if and only if

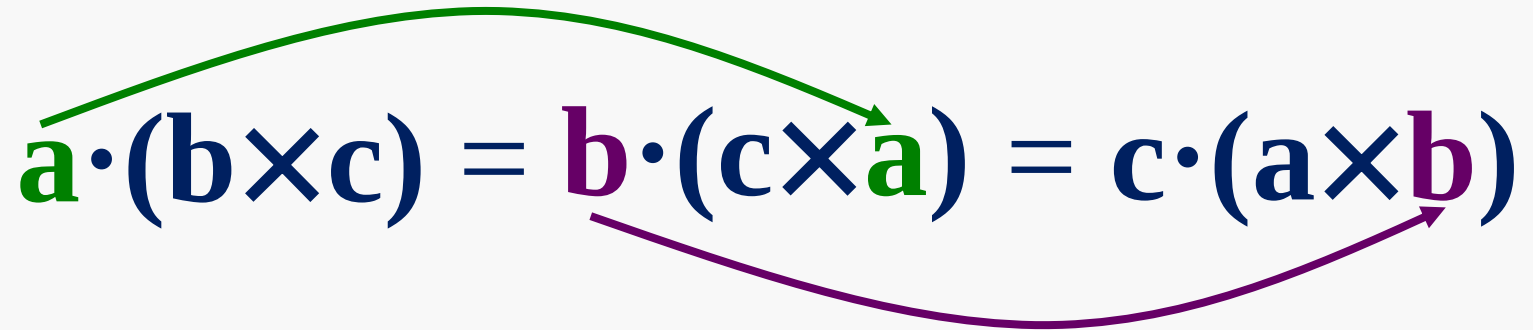
$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = 0$$

Coplanar (Cont')

Example 1.18: Show that the vectors $\mathbf{a} = \langle 1, 4, -7 \rangle$, $\mathbf{b} = \langle 2, -1, 4 \rangle$ and $\mathbf{c} = \langle 0, -9, 18 \rangle$ are coplanar; that is they lie on the same plane.

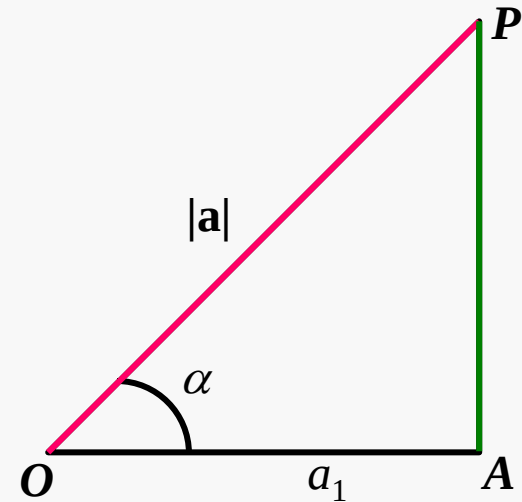
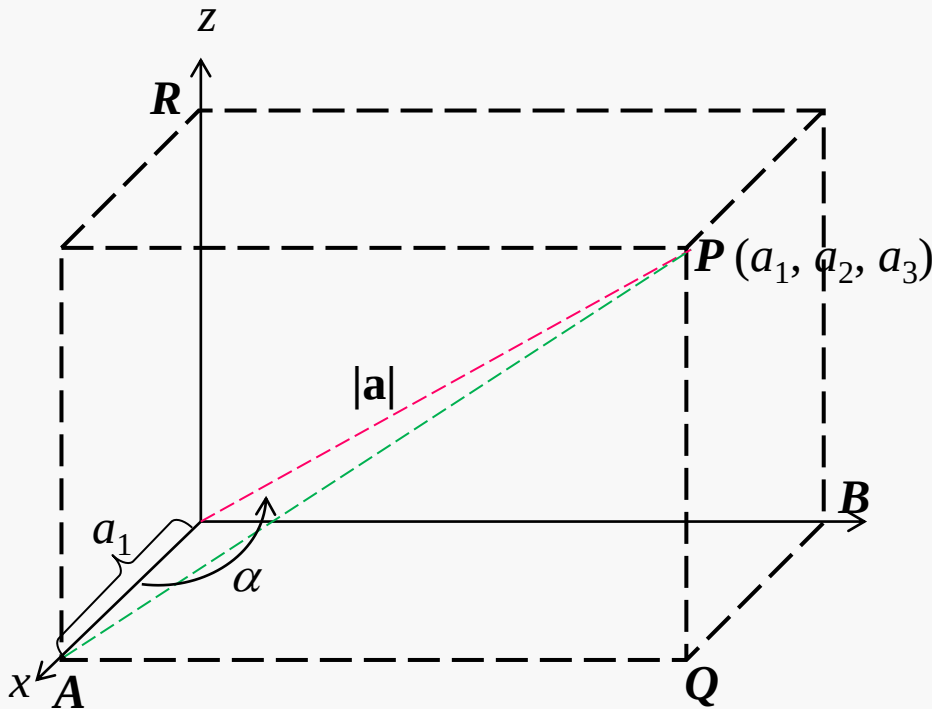
$$\begin{aligned}\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) &= \begin{vmatrix} 1 & 4 & -7 \\ 2 & -1 & 4 \\ 0 & -9 & 18 \end{vmatrix} = 1 \begin{vmatrix} -1 & 4 \\ -9 & 18 \end{vmatrix} - 4 \begin{vmatrix} 2 & 4 \\ 0 & 18 \end{vmatrix} - 7 \begin{vmatrix} 2 & -1 \\ 0 & -9 \end{vmatrix} \\ &= 1(18) - 4(36) - 7(-18) \\ &= 0\end{aligned}$$

Scalar Triple product

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \mathbf{b} \cdot (\mathbf{c} \times \mathbf{a}) = \mathbf{c} \cdot (\mathbf{a} \times \mathbf{b})$$


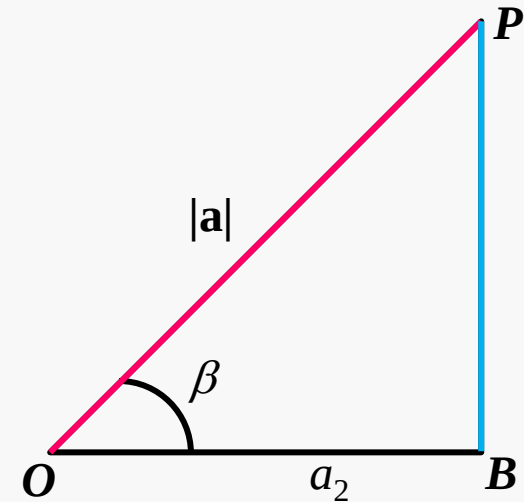
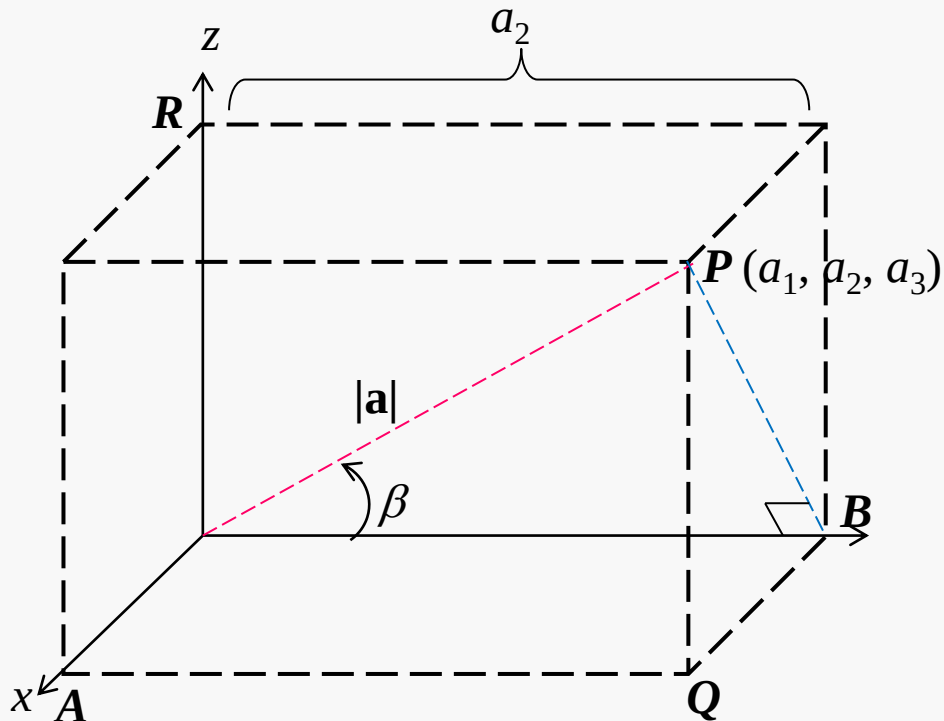
Direction angles and cosines

- The direction angles of a nonzero vector $\mathbf{a}$ are the angles α , β , and γ (in the interval $[0, \pi]$) that $\mathbf{a}$ makes with the position x -, y -, and z -axes.



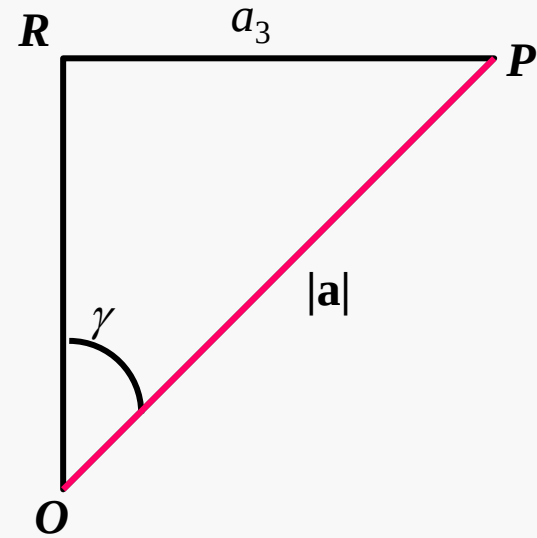
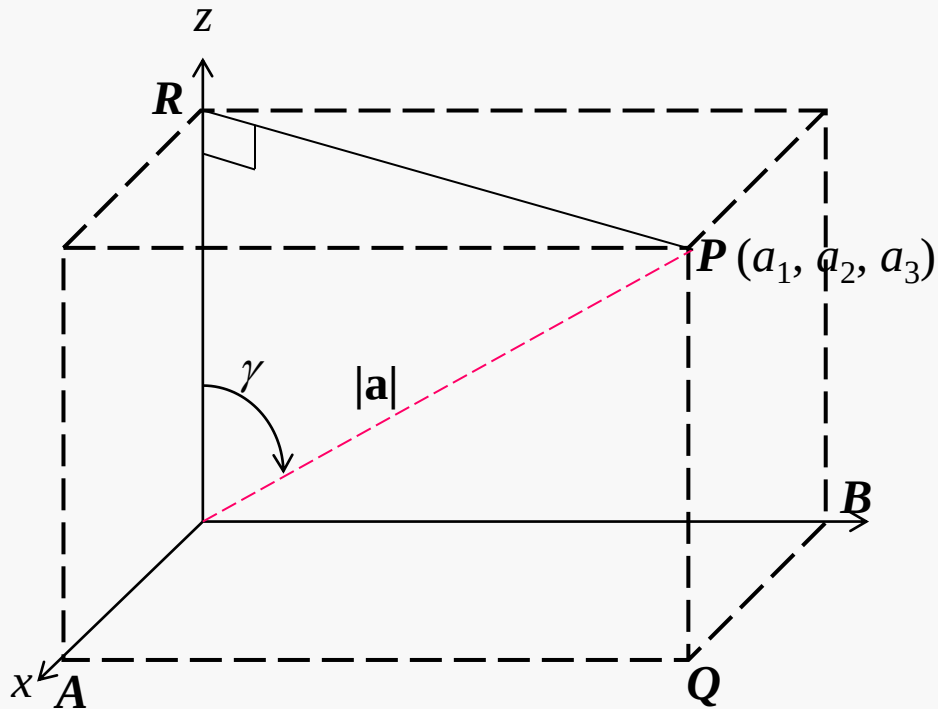
$$\cos \alpha = \frac{a_1}{\|\mathbf{a}\|} = \frac{a_1}{\sqrt{a_1^2 + a_2^2 + a_3^2}}$$

Direction angles and cosines (Cont')



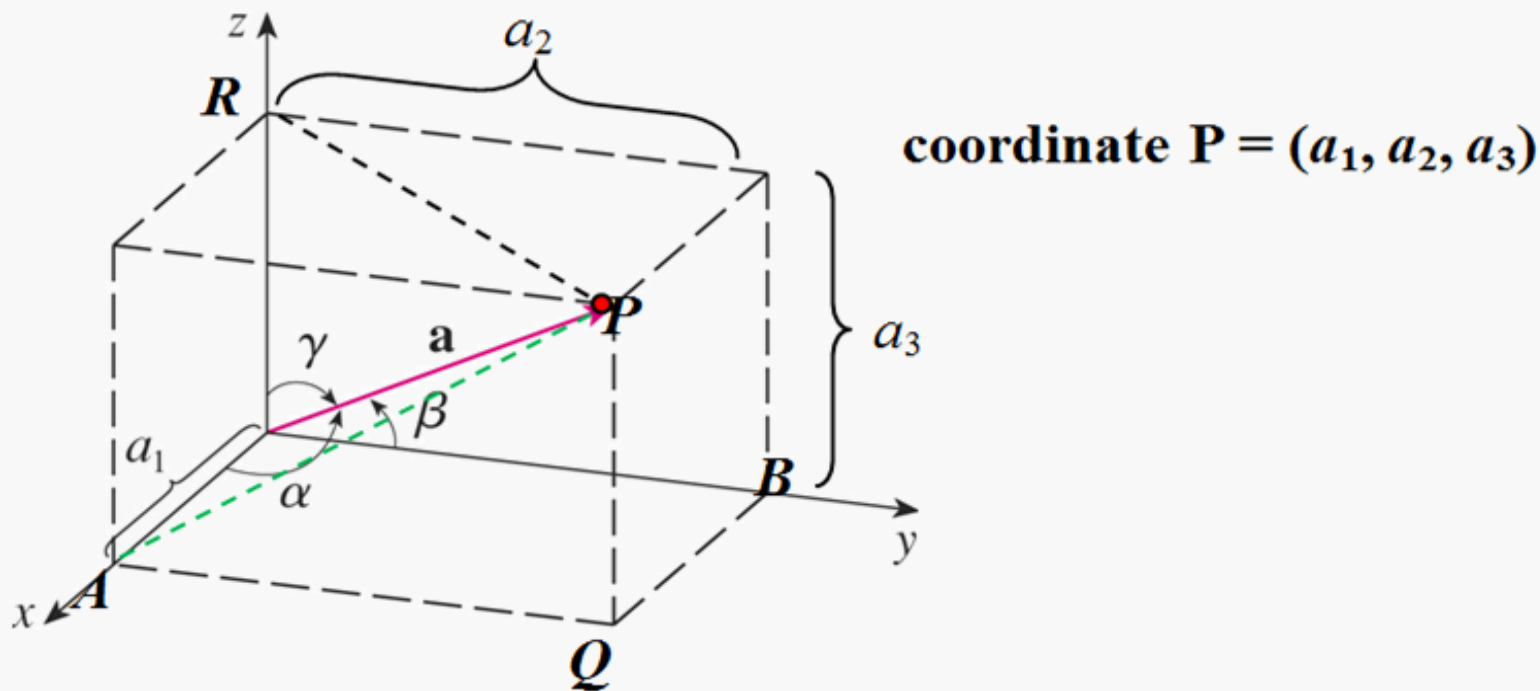
$$\cos \beta = \frac{a_2}{\|\mathbf{a}\|} = \frac{a_2}{\sqrt{a_1^2 + a_2^2 + a_3^2}}$$

Direction angles and cosines (Cont')



$$\cos \gamma = \frac{a_3}{\|\mathbf{a}\|} = \frac{a_3}{\sqrt{a_1^2 + a_2^2 + a_3^2}}$$

Direction angles and cosines (Cont')



$\cos \alpha$, $\cos \beta$, $\cos \gamma$ are called the direction cosines of $\mathbf{a}$.

The direction angle of $\mathbf{a}$ are the angle α , β , γ ($0 \leq \alpha, \beta, \gamma \leq \pi$) between $\mathbf{a}$ and the x , y , z - axes.

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

The ratio $a_1 : a_2 : a_3$ is called the direction ratio for vector $\mathbf{a}$.

Direction angles and cosines (Cont')

Example 1.19:

Determine the direction cosines and direction angles
 $\mathbf{a} = \langle 2, 1, -4 \rangle$.

$$\|\mathbf{a}\| = \sqrt{2^2 + 1^2 + (-4)^2} = \sqrt{4 + 1 + 16} = \sqrt{21}$$

The direction cosines and angles are :

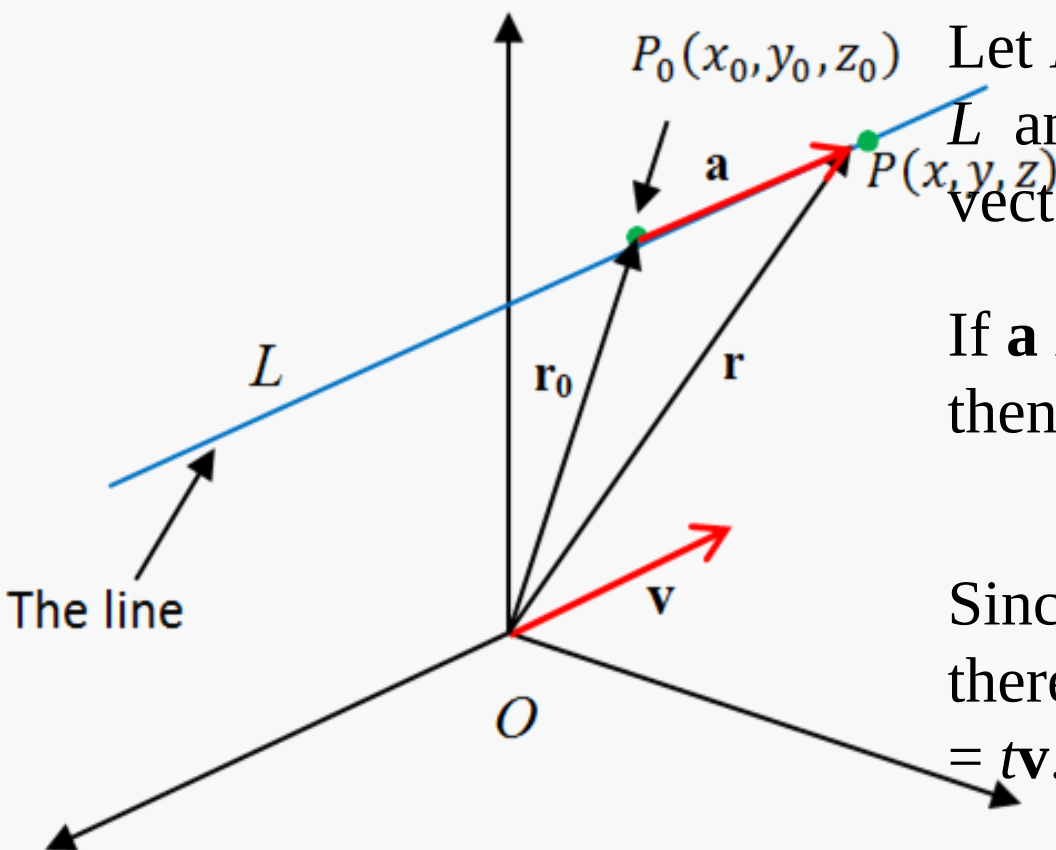
$$\cos \alpha = \frac{2}{\sqrt{21}} \qquad \cos \beta = \frac{1}{\sqrt{21}} \qquad \cos \gamma = \frac{-4}{\sqrt{21}}$$

$$\alpha = 1.119 \text{ rad or } 64.1^\circ$$

$$\beta = 1.351 \text{ rad or } 77.4^\circ$$

$$\gamma = 2.632 \text{ rad or } 150.8^\circ$$

Vector equation of a line



Let $P(x, y, z)$ be an arbitrary point on L and let $\mathbf{r}_0$ and $\mathbf{r}$ be the position vectors of P_0 and P

If $\mathbf{a}$ is the vector representation $\overrightarrow{P_0P}$, then

$$\mathbf{r} = \mathbf{r}_0 + \mathbf{a}$$

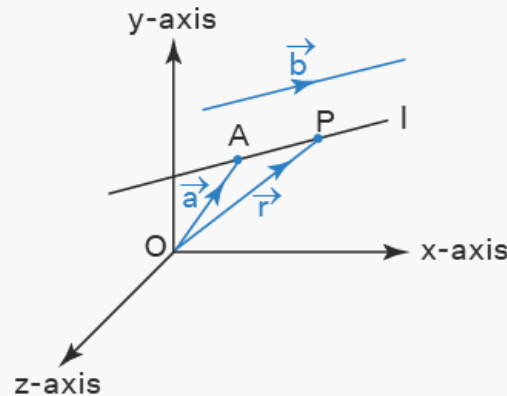
Since $\mathbf{a}$ and $\mathbf{v}$ are parallel vectors, there is a scalar t ($t \in \mathbb{R}$) such that $\mathbf{a} = t\mathbf{v}$. Thus

$$\mathbf{r} = \mathbf{r}_0 + t\mathbf{v}$$

which is vector equation of line L .

Vector equation of a line (Cont')

- Vector equations of a line can be computed with the help of any two points on the line, or with the help of a point on the line and a parallel vector.
- The two methods of forming a vector form of the equation of a line are as follows.

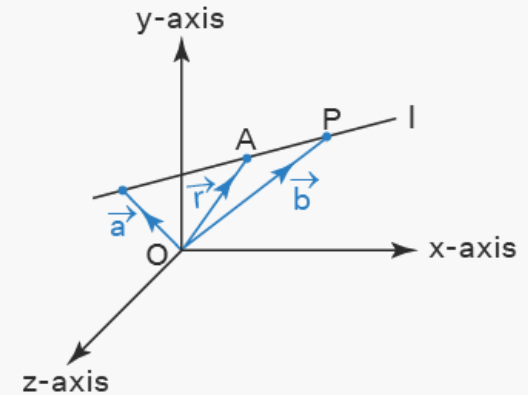


$$\overrightarrow{OA} = \vec{a}$$

$$\overrightarrow{OP} = \vec{r}$$

$$\vec{b} \parallel l$$

$$\vec{r} = \vec{a} + \lambda \vec{b}$$



$$\overrightarrow{OA} = \vec{a}$$

$$\overrightarrow{OP} = \vec{r}$$

$$\overrightarrow{OB} = \vec{b}$$

$$\vec{r} = \vec{a} + \lambda (\vec{b} - \vec{a})$$

- The vector equation of a line passing through a point and having a position vector $\vec{a}$, and parallel to a vector line $\vec{b}$ is $\vec{r} = \vec{a} + \lambda \vec{b}$.
- The vector equation of a line passing through two points with the position vector $\vec{a}$, and $\vec{b}$ is $\vec{r} = \vec{a} + \lambda (\vec{b} - \vec{a})$.

Vector equation of a line (Cont')

Let $\mathbf{v} = \langle a, b, c \rangle$, $\mathbf{r} = \langle x, y, z \rangle$, $\mathbf{r}_0 = \langle x_0, y_0, z_0 \rangle$

$$\mathbf{r} = \mathbf{r}_0 + t\mathbf{v}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix} + t \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} x_0 + ta \\ y_0 + tb \\ z_0 + tc \end{bmatrix}$$

Parametric equations:

$$x = x_0 + ta$$

$$y = y_0 + tb$$

$$z = z_0 + tc$$

Cartesian/**Symmetric** equations:

$$t = \frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

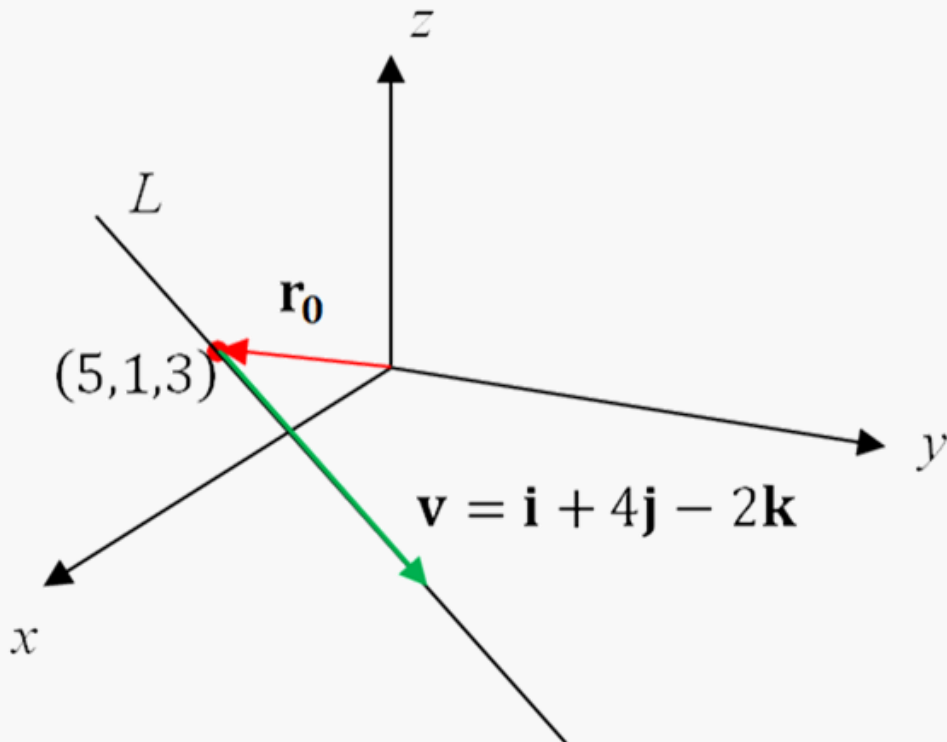
Vector equation of a line (Cont')

Example 1.20:

- (a) Find a vector equation and parametric equations for the line that passes through the point $(5, 1, 3)$ and is parallel to the vector $\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$.

$$\mathbf{r}_0 = \langle 5, 1, 3 \rangle$$

$$\mathbf{v} = \langle 1, 4, -2 \rangle$$



$$L = \begin{bmatrix} 5 \\ 1 \\ 3 \end{bmatrix} + t \begin{bmatrix} 1 \\ 4 \\ -2 \end{bmatrix}$$

Vector equation of a line (Cont')

(b) Find two other points on the line.

$$L = \begin{bmatrix} 5 \\ 1 \\ 3 \end{bmatrix} + t \begin{bmatrix} 1 \\ 4 \\ -2 \end{bmatrix}$$

Choosing the parameter value $t = 1$ gives

$$x = 6, y = 5, \text{ and } z = 1,$$

so $(6, 5, 1)$ is a point on the line.

Similarly, $t = -1$, gives the point $(4, -3, 5)$.

Vector equation of a line (Cont')

Note:

- Vector equation and parametric equations of a line are **not unique**.
- If we change the initial point or the parameter or choose a different parallel vector, then the equations changed.
- If $a = 0$, $x = x_0$, $\frac{y - y_0}{b} = \frac{z - z_0}{c}$ this means that L lies in the vertical plane $x = x_0$.

Vector equation of a line (Cont')

Example 1.21: (*Equations of a line through two points*)

- (a) Find parametric equations and Cartesian equations of the line that passes through the points $A(2, 4, -3)$ and $B(3, -1, 1)$.

$$\mathbf{v} = \overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB} = \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix} - \begin{bmatrix} 2 \\ 4 \\ -3 \end{bmatrix} = \begin{bmatrix} 1 \\ -5 \\ 4 \end{bmatrix} \quad \therefore L = \begin{bmatrix} 2 \\ 4 \\ -3 \end{bmatrix} + t \begin{bmatrix} 1 \\ -5 \\ 4 \end{bmatrix}$$

Parametric equations:

$$x = 2 + t \qquad y = 4 - 5t \qquad z = -3 + 4t$$

Cartesian equations:

(also called symmetric equation)

$$\frac{x-2}{1} = \frac{y-4}{-5} = \frac{z+3}{4}$$

Vector equation of a line (Cont')

Example 1.21(Cont')

(b) At what point does this line intersect the xy -plane?

Intersect on xy -plane when $z = 0$

$$\frac{x-2}{1} = \frac{y-4}{-5} = \frac{z+3}{4}$$

$$\frac{x-2}{1} = \frac{y-4}{-5} = \frac{0+3}{4} = \frac{3}{4}$$

$$x = 2 + \frac{3}{4} = \frac{11}{4} \quad y = (-5)\frac{3}{4} + 4 = \frac{1}{4}$$

Line intersects the xy -plane at the point $\left(\frac{11}{4}, \frac{1}{4}, 0\right)$.

Vector equation of a line (Cont')

Example 1.22:

Show that the lines L_1 and L_2 with parametric equations

$$\begin{array}{llll} x = 1 + t & y = -2 + 3t & z = 4 - t & \text{and} \\ x = 2s & y = 3 + s & z = -3 + 4s & \text{are skew lines.} \end{array}$$

If L_1 and L_2 had a point of intersection, there would be values of t and s such that

$$1 + t = 2s \qquad -2 + 3t = 3 + s \qquad 4 - t = -3 + 4s$$

Solve the first two equations, get $t = \frac{11}{5}$ and $s = \frac{8}{5}$, and these values don't satisfy the third equation.

Therefore there are no values of t and s that satisfy the three equations, so L_1 and L_2 do not intersect. Thus L_1 and L_2 are skew lines.

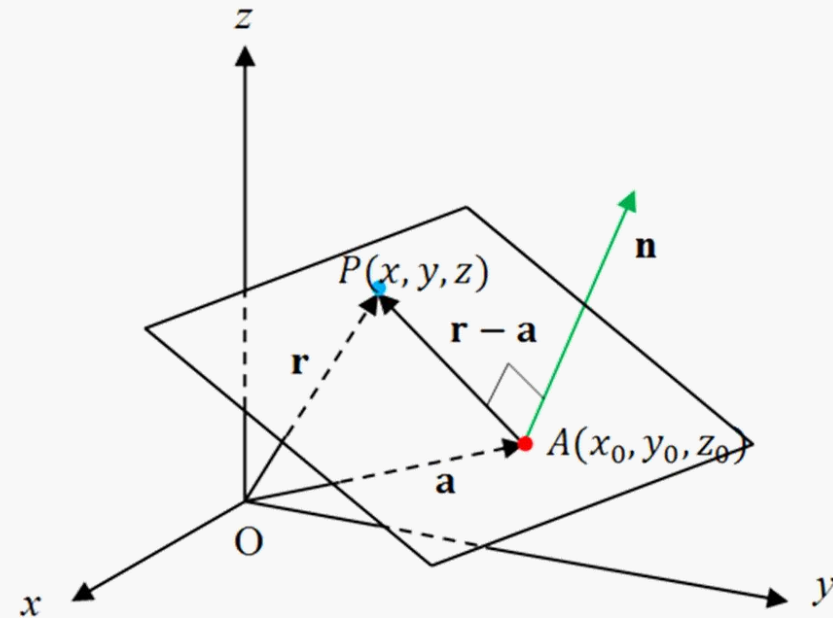
Vector equation of a plane

- To obtain the equation of a plane, we use the result that the **line joining any two points in the plane is perpendicular to the normal to the plane** as illustrated in figure.
- The vector $\mathbf{n}$ is perpendicular to the plane, $\mathbf{a}$ is the position vector of given point A in the plane and $\mathbf{r}$ is the position vector of any point P on the plane. The vector $\mathbf{r} - \mathbf{a}$ is perpendicular to $\mathbf{n}$, and hence

$$(\mathbf{r} - \mathbf{a}) \cdot \mathbf{n} = 0$$

- Which can write as

$$\mathbf{r} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n}$$



Vector equation of a plane(Cont')

$$\overrightarrow{AP} = \mathbf{r} - \mathbf{a} \perp \mathbf{n}$$

$$\text{so } (\mathbf{r} - \mathbf{a}) \cdot \mathbf{n} = 0$$

$$\mathbf{r} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n}$$

Let $\mathbf{n} = \langle a, b, c \rangle$, $\mathbf{r} = \langle x, y, z \rangle$ and $\mathbf{a} = \langle x_0, y_0, z_0 \rangle$

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

This equation is the **scalar equation of the plane through $A(x_0, y_0, z_0)$ with normal vector $\mathbf{n} = \langle a, b, c \rangle$.**

Vector equation of a plane(Cont')

Example 1.23:

Find an equation of the plane through the point (2, 4, -1) with normal vector $\mathbf{n} = \langle 2, 3, 4 \rangle$.

$$\mathbf{r} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$$

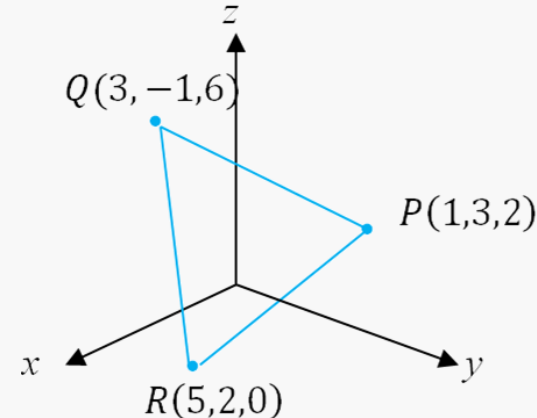
$$2x + 3y + 4z = 4 + 12 - 4 = 12$$

or use $a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$

$$2(x - 2) + 3(y - 4) + 4(z + 1) = 0$$

$$2x + 3y + 4z - 12 = 0$$

Vector equation of a plane(Cont')



Example 1.24:

Find an equation of the plane that passed through the points $P(1, 3, 2)$, $Q(3, -1, 6)$ and $R(5, 2, 0)$.

$$\overrightarrow{PQ} = \begin{bmatrix} 3-1 \\ -1-3 \\ 6-2 \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \\ 4 \end{bmatrix} \quad \overrightarrow{PR} = \begin{bmatrix} 5-1 \\ 2-3 \\ 0-2 \end{bmatrix} = \begin{bmatrix} 4 \\ -1 \\ -2 \end{bmatrix}$$

The cross product $\mathbf{PQ} \times \mathbf{PR}$ is orthogonal to the plane and can be taken as the normal vector

$$\mathbf{n} = \overrightarrow{PQ} \times \overrightarrow{PR} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -4 & 4 \\ 4 & -1 & -2 \end{vmatrix} = \begin{bmatrix} 12 \\ 20 \\ 14 \end{bmatrix}$$

use $\mathbf{r} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n}$

Or $a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$

$$12(x - 1) + 20(y - 3) + 14(z - 2) = 0$$

$$6x + 10y + 7z = 50$$

Vector equation of a plane(Cont')

Example 1.25:

Find the point at which the line with parametric equations $x = 2 + 3t$, $y = -4t$, $z = 5 + t$ intersects the plane $4x + 5y - 2z = 18$.

$$4(2 + 3t) + 5(-4t) - 2(5 + t) = 18$$

$$-10t = 20$$

$$t = -2$$

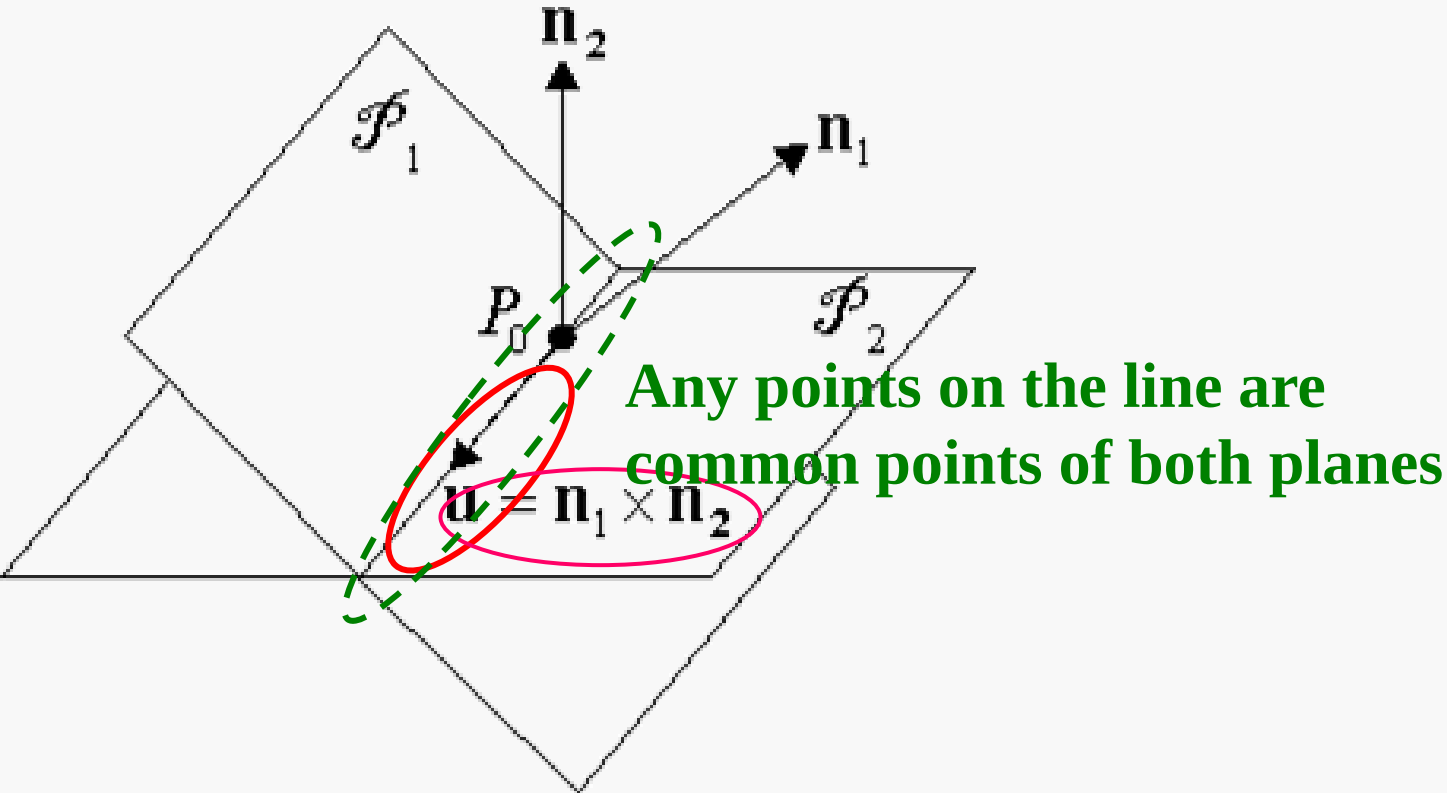
$$\text{when } t = -2, x = 2 + 3(-2) = -4$$

$$y = (-4)(-2) = 8$$

$$z = 5 + (-2) = 3$$

Hence, the point of intersection is $(-4, 8, 3)$.

Intersecting Line between Planes



- In 3D, two planes $\mathbf{P}_1$ and $\mathbf{P}_2$ are either parallel or they intersect in a single straight line $\mathbf{L}$.
- $\mathbf{u} = \mathbf{n}_1 \times \mathbf{n}_2$ is a direction vector for the intersection line $\mathbf{L}$ since $\mathbf{u}$ is perpendicular to both $\mathbf{n}_1$ and $\mathbf{n}_2$.
- two distinct planes intersect, then the set of common points is a line that lies in both planes.
- One of the common points of both planes is also a point on the line.
- For common point, we need to solve the two equations of the planes simultaneously.

Intersecting Line between Planes(Cont')

- To find the line of intersection of two planes we calculate the vector product (cross product) of the 2 planes' normals. This gives us the direction vector of the line. We then find a point on the line by letting $z = 0$ in the Cartesian equation of the two planes' equations and solving simultaneously for x and y
- Once we have both:
 - a direction vector of the line
 - a point on the line

we can write its vector equation.

Intersecting Line between Planes(Cont')

Example 1.26:

Write the parametric equations of the line of intersection of the planes $-5x + 11y + z = 34$ and $-2x + 4y + z = 9$.

Plane 1: $\mathbf{r} \cdot \begin{bmatrix} -5 \\ 11 \\ 1 \end{bmatrix} = 34$

Plane 2: $\mathbf{r} \cdot \begin{bmatrix} -2 \\ 4 \\ 1 \end{bmatrix} = 9$

Direction of the line: $\begin{bmatrix} -5 \\ 11 \\ 1 \end{bmatrix} \times \begin{bmatrix} -2 \\ 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 11-4 \\ -2-(-5) \\ -20-(-22) \end{bmatrix} = \begin{bmatrix} 7 \\ 3 \\ 2 \end{bmatrix}$

Any common point, set $y = 0$: $\left. \begin{array}{l} -5x + z = 34 \\ -2x + z = 9 \end{array} \right\} \begin{array}{l} \text{Solve simultaneously,} \\ \text{Common point } \left(-\frac{25}{3}, 0, -\frac{23}{3} \right) \end{array}$

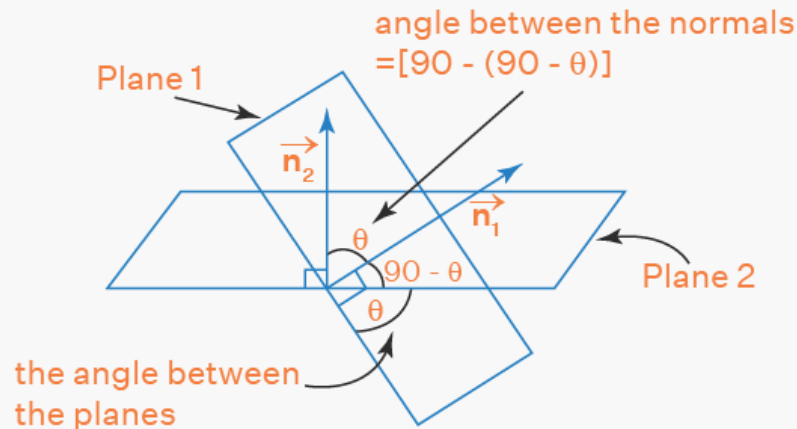
$\therefore$ Intersecting line,

$$\underset{\sim}{r} = \underset{\sim}{a} + \lambda \underset{\sim}{b} = \begin{bmatrix} -25/3 \\ 0 \\ -23/3 \end{bmatrix} + \lambda \begin{bmatrix} 7 \\ 3 \\ 2 \end{bmatrix}$$

What is Angle Between Two Planes?

- The angle between two planes is equal to the angle between the normal vectors to the two planes.
- We can determine the angle between two planes using the cartesian equation of the plane and the vector equation of the plane.
- Since the angle between the two planes is given by the angle between the normals to these two planes, therefore we use the scalar product and magnitudes of the normal vectors in the formula to find the angle between them.
- In vector form, the equation of a plane is given by $\mathbf{r} \cdot \mathbf{n} = d$, and its cartesian equation is given by, $Ax + By + Cz + D = 0$.
- Now, let us go through the formulas to find the angle between two planes.

Angle between planes Formula



- Now, there are two formulas to find the angle between two planes. The formulas exist in vector form and cartesian form. Consider two planes P_1 and P_2 and the angle between them is θ . The equations of the two planes in vector form are $\mathbf{r} \cdot \mathbf{n}_1 = d_1$ and $\mathbf{r} \cdot \mathbf{n}_2 = d_2$ and the equations of the two planes in the cartesian form are $A_1x + B_1y + C_1z + D_1 = 0$ and $A_2x + B_2y + C_2z + D_2 = 0$. Then, the formulas to find the angle between two planes are:
- $\cos \theta = |(\mathbf{n}_1 \cdot \mathbf{n}_2)| / (|\mathbf{n}_1| \cdot |\mathbf{n}_2|)$
- $\cos \theta = |(A_1A_2 + B_1B_2 + C_1C_2)| / [\sqrt{(A_1^2 + B_1^2 + C_1^2)} \sqrt{(A_2^2 + B_2^2 + C_2^2)}]$
- Using the above formulas, we can determine the value of $\cos \theta$ and take $\cos$ inverse on both sides to find the value of θ , and hence, the angle between two planes.

Angle between planes Formula

For planes $\vec{r} \cdot \vec{n}_1 = d_1$ and $\vec{r} \cdot \vec{n}_2 = d_2$ and acute angle θ is,

$$\cos \theta = \left| \frac{\vec{n}_1 \cdot \vec{n}_2}{|\vec{n}_1| |\vec{n}_2|} \right|$$

For planes $A_1x + B_1y + C_1z + D_1 = 0$ and $A_2x + B_2y + C_2z + D_2 = 0$ acute angle θ is,

$$\cos \theta = \frac{|A_1A_2 + B_1B_2 + C_1C_2|}{\sqrt{A_1^2 + B_1^2 + C_1^2} \sqrt{A_2^2 + B_2^2 + C_2^2}}$$

- Let us now solve an example based on the formula of the angle between two planes in vector form. For planes, $\vec{r} \cdot \vec{n}_1 = d_1$ and $\vec{r} \cdot \vec{n}_2 = d_2$, we will use the formula $\cos \theta = |(\vec{n}_1 \cdot \vec{n}_2)| / (|\vec{n}_1| |\vec{n}_2|)$, where $\vec{n}_1$, and $\vec{n}_2$ are normal vectors to the two planes and θ is the angle between the two planes

Angle between planes

Example 1.27 :

(a) Find the angle between the planes

$$x + y + z = 1 \text{ and } x - 2y + 3z = 1.$$

Apply dot product rules, $\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta$

$$\cos \theta = \frac{\mathbf{n}_1 \cdot \mathbf{n}_2}{\|\mathbf{n}_1\| \|\mathbf{n}_2\|} = \frac{1(1) + 1(-2) + 1(3)}{\sqrt{1+1+1} \sqrt{1+4+9}} = \frac{2}{\sqrt{42}}$$

$$\theta = \cos^{-1} \left(\frac{2}{\sqrt{42}} \right) \approx 72^\circ$$

Angle between planes (Cont')

Example 1.27: (Cont')

any 2D plane

(b) Find symmetric equations for the line of intersection L of these two planes. $[x + y + z = 1 \text{ and } x - 2y + 3z = 1]$

First find a point where the line intersects the xy -plane $\Rightarrow$ set $z = 0$

$\Rightarrow x + y = 1$ and $x - 2y = 1$, solve simultaneously, obtain $x = 1, y = 0$, so 1 of the common points lies on L is $(1, 0, 0)$.

since L lies on both planes, it is **perpendicular** to both of the normal vectors.

$$\therefore L \text{ parallel to } \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \times \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ -3 \end{bmatrix}$$

$$\frac{x-1}{5} = \frac{y}{-2} = \frac{z}{-3}$$

$\therefore$ Symmetric equation of L :

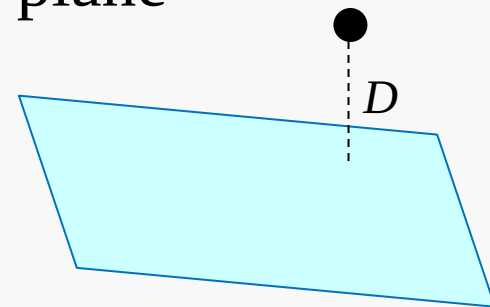
Distance from a point to a plane

- The distance between point and plane is the length of the perpendicular to the plane passing through the given point.
- In other words, we can say that the distance between point and plane is the length of the normal vector dropped from the given point onto the given plane.
- If we want to determine the distance between the point P with coordinates (x_0, y_0, z_0) and the given plane with equation $Ax + By + Cz = D$, then the distance between point P and the given plane is given by $|Ax_0 + By_0 + Cz_0 - D|/\sqrt{A^2 + B^2 + C^2}$.

Distance from a point to a plane

Distance D from a point $P_0(x_0, y_0, z_0)$ to the plane $ax + by + cz + d = 0$ is given as

$$D = \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}}$$



Example 1.28:

Find the distance D between the point $(1, -4, -3)$ and the plane $2x - 3y + 6z = -1$.

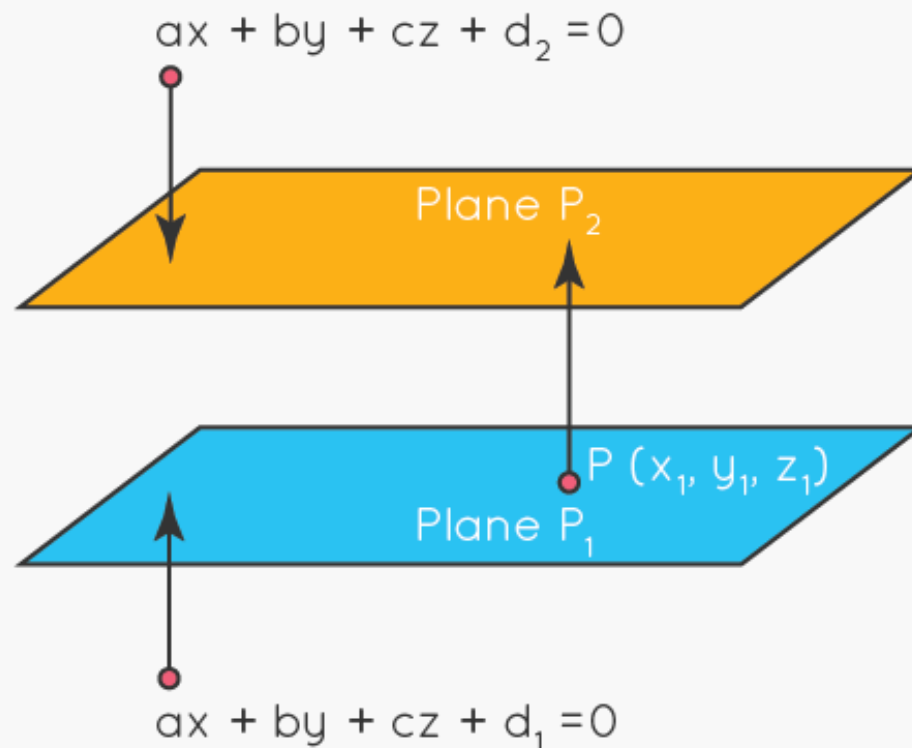
$$D = \frac{|2(1) + (-3)(-4) + 6(-3) + 1|}{\sqrt{2^2 + (-3)^2 + 6^2}} = \frac{|-3|}{7} = \frac{3}{7}$$

Distance between two parallel planes

- The formula for the distance between two parallel planes is similar to the formula to determine the distance between two parallel lines.
- As we know, the coordinates of the normal vectors of the two parallel planes are either proportional or equal.
- We will explore the method to determine the distance between the two planes using the point-plane distance formula.
- The formula to calculate the distance between a point (x_1, y_1, z_1) to a plane $ax + by + cz + d = 0$ is given by $D = \frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}$.
- To find the distance between two planes using the point-plane distance formula, we can follow the steps given below:

Distance between two parallel planes

- Step 1: Convert the equations of the two planes into the standard format, i.e., $ax + by + cz + d = 0$
- Step 2: Check if the planes are parallel. [Two planes P1: $a_1x + b_1y + c_1z + d_1 = 0$ and P2: $a_2x + b_2y + c_2z + d_2 = 0$ are parallel if $a_1/a_2 = b_1/b_2 = c_1/c_2$]
- Step 3: Consider the coefficients a, b, c, d from the equation of one of the planes.
- Step 4: Consider a point $P(x_1, y_1, z_1)$ on the other plane. [An easy way to find the point is by taking $x = y = 0$, and finding the value of z from the equation of the other plane]
- Step 5: Substitute the values of $a, b, c, d, x_1, y_1, z_1$ into the formula for the distance between a point and plane:
$$\frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}$$
- Following the above steps, we can find the distance between two planes using the formula for distance between point and plane.



To find the distance between two parallel planes, one can use the formula:

$$D = \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}}$$

where (x_0, y_0, z_0) is a point in one of the planes, and the equation of the other plane is of the form $ax + by + cz + d = 0$

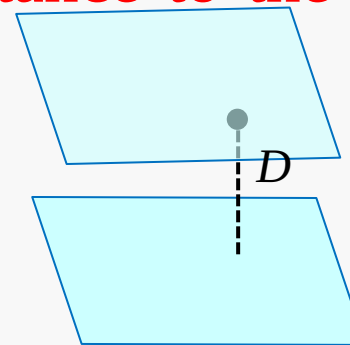
Distance between two parallel planes

Example 1.29:

Find the distance between the parallel planes $10x + 2y - 2z = 5$ and $5x + y - z = 1$.

Choose any point on one plane and calculate its distance to the other plane.

Choose any point from (first) plane $10x + 2y - 2z = 5$,
let $y = z = 0$, $10x + 0 - 0 = 5$, $x = 0.5$



Then use (Second) plane $5x + y - z = 1$ to calculate the distance.

$\therefore$ Find the distance D between the point $(0.5, 0, 0)$ and the plane $5x + y - z = 1$.

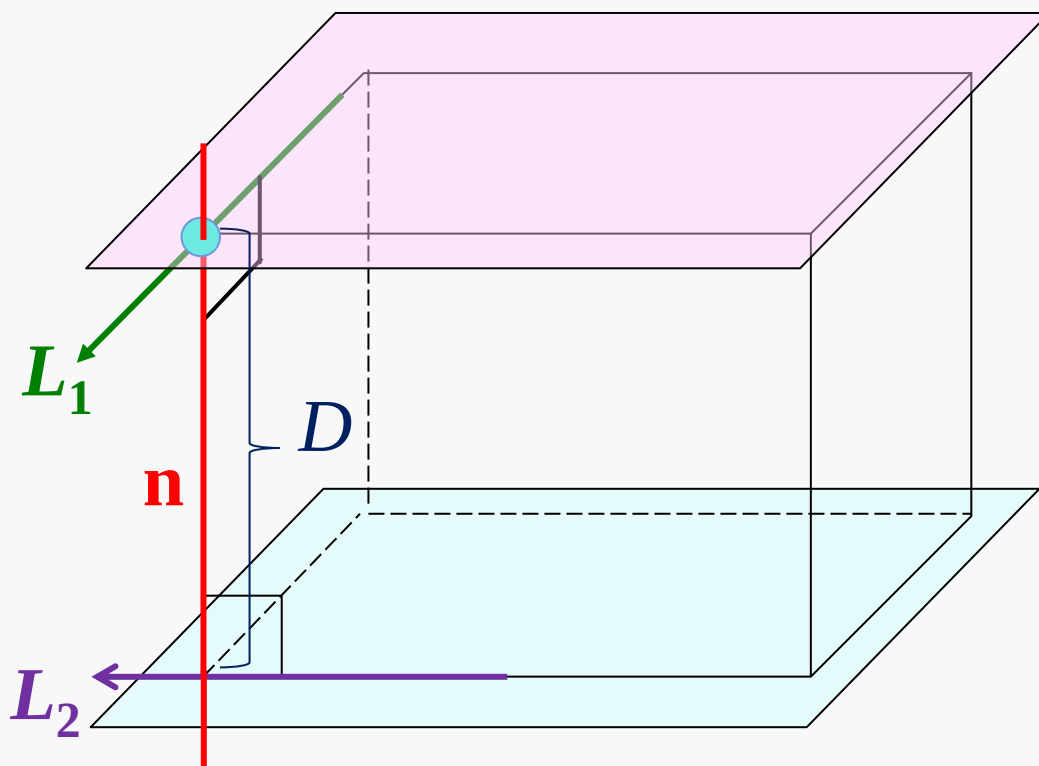
$$D = \frac{\left| 5\left(\frac{1}{2}\right) + 1(0) + (-1)(0) - 1 \right|}{\sqrt{5^2 + 1^2 + (-1)^2}} = \frac{\sqrt{3}}{6}$$

Distance between two skew lines

In previous example 1.22, we showed that the lines

$$\begin{array}{lll} L_1: x = 1 + t & y = -2 + 3t & z = 4 - t \\ L_2: x = 2s & y = 3 + s & z = -3 + 4s \end{array}$$

are skew. Find the distance between them.



Distance between two skew lines(Cont')

In previous example, we showed that the lines

$$\begin{aligned} L_1: x &= 1 + t & y &= -2 + 3t & z &= 4 - t \\ L_2: x &= 2s & y &= 3 + s & z &= -3 + 4s \end{aligned} \quad L_1 = \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix} + t \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix} \quad L_2 = \begin{bmatrix} 0 \\ 3 \\ -3 \end{bmatrix} + s \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix}$$

are skew. Find the distance between them.

① Find common perpendicular vector,

$$\mathbf{n} = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix} \times \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 13 \\ -6 \\ -5 \end{bmatrix}$$

② Choose a point from L_2 (or L_1), form a plane equation

Let $s = 0$, obtain point $(0, 3, -3)$, apply $\mathbf{r} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n}$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} \cdot \begin{bmatrix} 13 \\ -6 \\ -5 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \\ -3 \end{bmatrix} \cdot \begin{bmatrix} 13 \\ -6 \\ -5 \end{bmatrix}$$

Plane equation: $13x - 6y - 5z + 3 = 0$

③ Choose a point from L_1 , calculate the distance from the plane

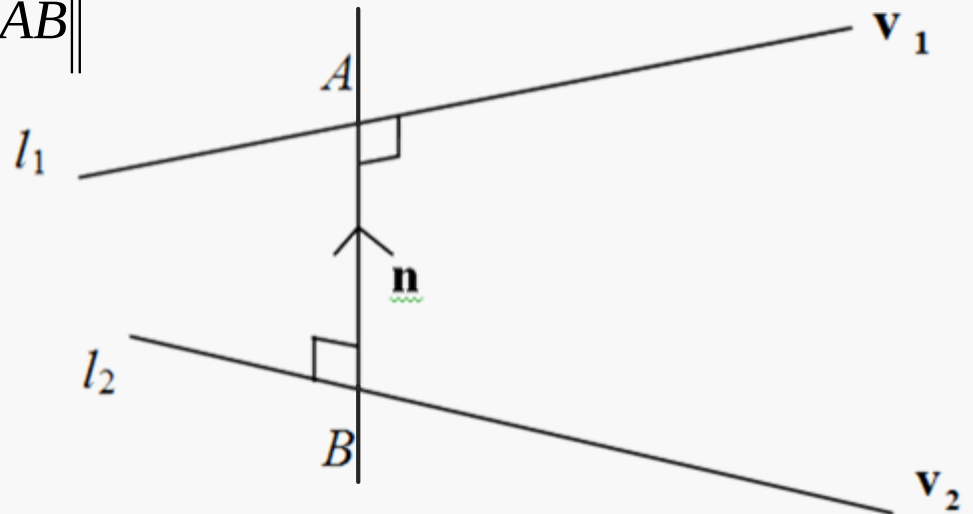
Let $t = 0$, obtain point $(1, -2, 4)$,

$$D = \frac{|13(1) + (-6)(-2) + (-5)(4) + 3|}{\sqrt{13^2 + (-6)^2 + (-5)^2}} = \frac{8}{\sqrt{230}} \approx 0.53$$

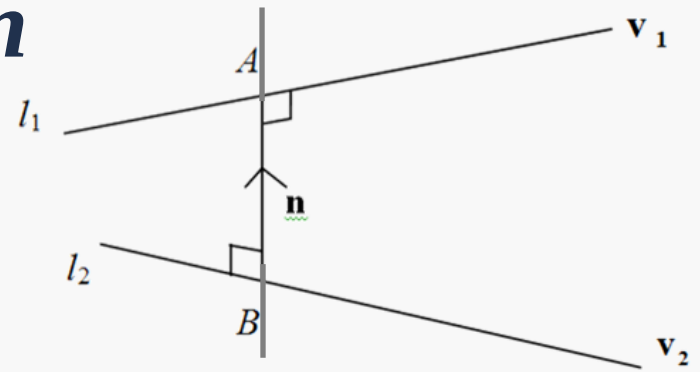
Shortest Distance between two skew lines

- (i) Find the cross product of the direction $\mathbf{v}_1$ and $\mathbf{v}_2$, $\vec{AB} = \mathbf{v}_1 \times \mathbf{v}_2$.
- (ii) Choose any points C and D , one on l_1 and l_2
- (iii) Find the length of the projection of CD on the vector found in (i) : a multiple of AB
- (iv) Shortest distance,

$$d = \frac{|\vec{CD} \cdot \vec{AB}|}{\|\vec{AB}\|}$$



Shortest Distance between two skew lines (Cont')



Find the shortest distance between two skew lines:

$$\mathbf{r}_1 = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} + s \begin{bmatrix} -9 \\ -2 \\ 2 \end{bmatrix} \quad \mathbf{r}_2 = \begin{bmatrix} 7 \\ 2 \\ -2 \end{bmatrix} + t \begin{bmatrix} -15 \\ -4 \\ 2 \end{bmatrix}$$

Common perpendicular vector, $\vec{AB} = \mathbf{n} = \begin{bmatrix} -9 \\ -2 \\ 2 \end{bmatrix} \times \begin{bmatrix} -15 \\ -4 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ -12 \\ 6 \end{bmatrix}$

$$\|\mathbf{n}\| = \sqrt{4^2 + (-12)^2 + 6^2} = \sqrt{196} = 14$$

Choose any points from $\mathbf{r}_1$ and $\mathbf{r}_2$:

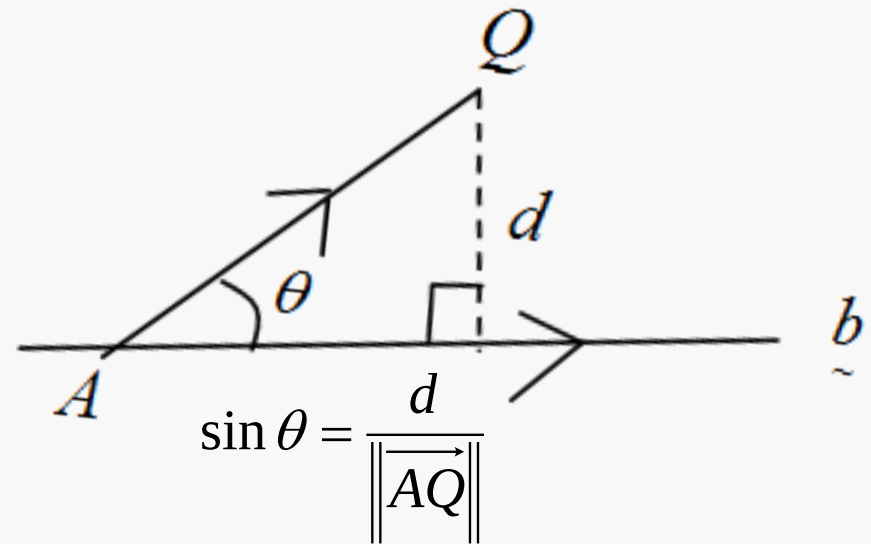
C (2, 1, 2) from $\mathbf{r}_1$ & D (7, 2, -2) from $\mathbf{r}_2$, $\vec{CD} = \begin{bmatrix} 5 \\ 1 \\ -4 \end{bmatrix}$

$$d = \frac{|\vec{CD} \cdot \vec{n}|}{\|\vec{AB}\|} = \frac{|\langle 5, 1, -4 \rangle \cdot \langle 4, -12, 6 \rangle|}{14} = \frac{|20 + (-12) + (-24)|}{14} = \frac{8}{7}$$

Shortest Distance from a point to line

By cross product rules:

$$\begin{aligned}\|\vec{AQ} \times \mathbf{b}\| &= \|\vec{AQ}\| \|\mathbf{b}\| \sin \theta \\ &= \|\vec{AQ}\| \|\mathbf{b}\| \frac{d}{\|\vec{AQ}\|} \\ d &= \frac{\|\vec{AQ} \times \mathbf{b}\|}{\|\mathbf{b}\|}\end{aligned}$$



Shortest Distance from a point to line

(Cont')

Example 1.32:

Find the shortest distance from the point $Q(1, 5, 1)$ to the line $\mathbf{r} = \langle 0, 5, 2 \rangle + t\langle 1, -2, 1 \rangle$.

$$\overrightarrow{AQ} = \begin{bmatrix} 1-0 \\ 5-5 \\ 1-2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \quad \overrightarrow{AQ} \times \mathbf{b} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \times \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0-2 \\ -1-1 \\ -2-0 \end{bmatrix} = \begin{bmatrix} -2 \\ -2 \\ -2 \end{bmatrix}$$

$$d = \frac{\|\overrightarrow{AQ} \times \mathbf{b}\|}{\|\mathbf{b}\|} = \frac{\sqrt{(-2)^2 + (-2)^2 + (-2)^2}}{\sqrt{1^2 + (-2)^2 + 1^2}} = \sqrt{\frac{12}{6}} = \sqrt{2}$$